Systematic methods for structurally consistent coarse-grained models

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ABSTRACT

The present chapter provides a primer on theories for coarse-grained (CG) modeling and, in particular, reviews several systematic methods for determining effective potentials for CG models. The chapter first reviews a statistical mechanics framework for relating atomistic and CG models. This framework naturally leads to a quantitative criterion for CG models that are "consistent" with a particular atomistic model for the same system. This consistency criterion is equivalent to minimizing the relative entropy between the two models. This criterion implies that a many-body PMF is the appropriate potential for a CG model that is consistent with a particular atomistic model. The chapter then presents a unified exposition of the theory and numerical methods for several approaches for approximating this many-body PMF. Finally, the chapter closes with a brief discussion of a few of the outstanding challenges facing the field of systematic coarse-graining.

1. INTRODUCTION

As clearly demonstrated by the previous chapters, classical molecular dynamics (MD) and Monte Carlo simulations of atomically detailed models have contributed profound insight into many biological and condensed phase systems. [1, 2] Classical force fields have been carefully parameterized to provide a reasonably realistic description of the atomistic structure and interactions for many molecular systems over a relatively wide range of thermodynamic conditions. [3–5] With current hardware [6, 7] and software [8–12] resources, atomistic simulation methods allow exquisitely detailed investigations of phenomena that evolve on nanometer length scales and perhaps even microsecond time scales. [13, 14] However, as a direct consequence of this detail, atomistic simulation methods remain limited in their capability for efficiently and exhaustively investigating many processes of fundamental importance that evolve on significantly longer length and time scales, e.g., the folding of multidomain proteins or signal transduction events. These limitations have motivated tremendous interest in "coarse-grained" computational models that provide greater efficiency by representing systems in somewhat reduced detail.

The notion of "coarse-graining" high resolution models to develop more tractable low resolution models for studying emergent phenomena of interest is a recurring theme throughout

science. [15] Hydrodynamics [16] and thermodynamics [17] are two obvious examples of such coarse-grained theories. Similarly, classical atomistic models are frequently interpreted as a coarse-grained description of more detailed quantum mechanical models. [18] Therefore, it is not surprising that pioneers such as Kirkwood made many seminal contributions to the statistical mechanics of coarse-graining long ago. [19–21] Nor is it surprising that a staggering diversity of coarse-grained models have been applied to study a vast array of physical and biological phenomena. These models span a wide range from continuum and field theoretic models to particle-based models. Consequently, the present introduction to coarse-grained modeling will necessarily be of relatively limited scope.

This chapter will address "mesoscopic" particle-based coarse-grained (CG) models that retain sufficient detail to allow for chemically important distinctions. The fundamental particles in such models are interaction sites (sometimes called super-atoms) that represent groups of atoms, typically on the order of 2-10 "heavy" atoms (e.g., C, N, O, P) and associated hydrogen atoms, although this chapter certainly also applies for "bigger" sites. The interactions between these sites are then described by effective potentials that incorporate both energetic and entropic contributions resulting from averaging over (or, more formally, integrating out) uninteresting atomic details. Simulations with the resulting models can be three to four orders of magnitude more efficient than atomically-detailed simulations for several reasons: Because CG models require fewer interacting sites to model the same system, fewer interactions must be calculated at each time step. Furthermore, these interactions are typically of shorter-range than the interactions in atomistic models, e.g., long-ranged Coulombic interactions may be more effectively screened on the length scales relevant for CG interactions and can be subsumed into short ranged effective potentials. These considerations are especially important for implicit solvent CG models that formally integrate out all of the solvent molecules and incorporate their effects into the interactions between the remaining sites. [22] Moreover, as a consequence of averaging over atomistic configurations, the CG potential can implicitly include entropic effects and greatly reduce the need to explicitly sample a large number of degenerate atomically detailed configurations. [23] Finally, by averaging over interactions, the interactions between CG sites may be "softer," (i.e., of lower frequency) which allows a larger integration time step for propagating dynamics. [24] This effect is further enhanced because the friction and memory effects associated with fluctuations on an atomically-detailed potential energy surface are dramatically reduced, allowing

the CG model to rapidly sample larger regions of configuration space. However, as a side effect of this averaging, the CG effective potential may not be transferable to multiple systems or thermodynamic state points and the relationship between the dynamics of atomistic and CG models becomes somewhat unclear.

Coarse-grained models have often been employed as intellectually appealing minimal models that are constructed to capture certain basic physical principles. These models then provide a powerful tool for assessing how generic design principles may govern phenomenon of interest, e.g., polymer configurations, [25, 26] curvature-mediated aggregation of membrane-bound proteins, [27] or protein folding. [28] However, such minimal models may not provide a realistic description of any particular system. [18] Moreover, these models may give the right answer for the wrong reason, as they only reflect the interactions that were initially specified in designing the model. [29] While CG models may allow highly efficient investigations of complex phenomena, the resulting models may be misleading unless they incorporate "the right physics." One possible way to quantify this statement would be to require that the model generate the correct distribution of CG microstates, i.e., the distribution of microstates that would be obtained if one used a "coarse-grained lens" to observe the configurations generated by an accurate high resolution model. The present chapter will consider this possibility in detail.

These considerations have motivated extensive recent effort in developing systematic "bottom-up" approaches for determining "chemically specific" CG models that generate results that are consistent with an accurate high resolution model for a particular system. These approaches typically focus on accurately reproducing either thermodynamic or structural properties. Klein and coworkers [30–36] and Marrink and coworkers [24, 37–41] have pioneered the development of CG models that reproduce thermodynamic properties determined from either experiment or simulation. In particular, surface tension, partitioning free energies, and density have frequently been employed to parameterize Lennard-Jones-type pair potentials between point sites. The resulting models have demonstrated remarkable transferability for various systems and a wide range of thermodynamic state points. In some cases, these models have also reproduced structural properties with semiquantitative accuracy. The recent developments by the Marrink group are reviewed in a later chapter.

"Bottom-up" strategies have also been developed for parameterizing CG potentials to accurately reproduce structural data obtained from experiments or atomistic simulations.

This chapter will focus on the theory and development of these approaches. The primary objectives of the present chapter are as follows:

1. Provide a readily accessible introduction to the statistical mechanical framework for relating atomistic and CG models of the same system.

2. Formulate and analyze quantitative consistency criterion for CG models that are "structurally consistent" with an underlying atomistic model.

3. Present a unified discussion of the computational and theoretical aspects of certain systematic methods for determining such models.

4. Outline some of the remaining outstanding challenges facing these approaches to systematic coarse-graining.

In particular, it is our hope that the present chapter will be distinguished from previous reviews by the accessibility of the introductory material and also by the unified approach to discussing a variety of systematic coarse-graining methods.

It is important to emphasize several major short-comings of this chapter. As noted above, the present chapter is highly limited in scope and reflects the notation and personal perspective of the author. Moreover, this chapter does not attempt to address the staggering diversity in scope, application, and approach of coarse-grained models. For more information, the reader is referred to a several recent themed journal issues, [42–45] and a recent book [46] devoted to the subject of CG models. In particular, many recent reviews [47–49] have addressed CG dynamics [50, 51] and models of, e.g., protein, [18, 52–61] lipid, [62–70] and polymer systems. [71–74] Furthermore, the remaining chapters of this book provide additional insight into the great diversity and utility of CG models for a wide variety of applications.

2. CONSISTENCY BETWEEN ATOMISTIC AND COARSE-GRAINED MODELS

This chapter considers consistency between high resolution and low resolution mechanical models of a particular molecular system in an equilibrium canonical ensemble. The high resolution model will be referred to as an "atomistic model" and the corresponding interacting

particles in the atomistic model will be referred to as "atoms." The low resolution model will be referred to as a "coarse-grained model" and the corresponding interacting particles in the coarse-grained (CG) model will be referred to as "sites." However, the following framework applies quite generally for parameterizing low resolution particle-based models from high resolution particle-based models. It will be assumed that neither the atomistic nor the CG model includes rigid constraints. We shall also primarily address the case that the atoms and the sites correspond to structureless point particles, although it is possible to consider sites corresponding to asymmetric mass distributions [75–78] or that include internal degrees of freedom. [79] The presentation and notation reflects the framework developed by Voth, Andersen, and coworkers. [80]

2.1. Models and Notation

We begin by introducing notation and establishing the elementary statistical thermodynamics of atomistic and CG models. [81] For convenience, atomistic quantities will be represented by lower case symbols, while CG quantities will be represented by capitalized symbols.

The microstate of an atomistic model with n atoms is specified by the set of n Cartesian coordinates $\mathbf{r} = (\mathbf{r}_1, \ldots, \mathbf{r}_n)$ and momenta $\mathbf{p} = (\mathbf{p}_1, \ldots, \mathbf{p}_n)$. The Hamiltonian for the atomistic model is:

$$h(\mathbf{r}, \mathbf{p}) = \sum_i \frac{1}{2m_i}\mathbf{p}_i^2 + u(\mathbf{r}), \tag{1}$$

where m_i is the mass of atom i and $u(\mathbf{r})$ is a classical interaction potential for the model. No assumptions will be made about the particular form of u. The canonical phase space distribution for the atomistic model factors into statistically independent configuration and momenta distributions:

$$p_r(\mathbf{r}) \propto \exp[-u(\mathbf{r})/k_B T] \tag{2}$$

$$p_p(\mathbf{p}) \propto \exp\left[-\sum_i \mathbf{p}_i^2/2m_i k_B T\right]. \tag{3}$$

In particular, the momenta distribution factors into independent Gaussian distributions for each $\mathbf{p}_i$.

The microstate of a CG model with N sites is specified by the set of N Cartesian coordinates $\mathbf{R} = (\mathbf{R}_1, \ldots, \mathbf{R}_N)$ and momenta $\mathbf{P} = (\mathbf{P}_1, \ldots, \mathbf{P}_N)$. The Hamiltonian for the CG

model is:

$$H(\mathbf{R},\mathbf{P}) = \sum_I \frac{1}{2M_I}\mathbf{P}_I^2 + U(\mathbf{R}), \tag{4}$$

where M_I is the mass of site I and $U(\mathbf{R})$ is a classical interaction potential for the model. The canonical phase space distribution for the CG model factors into statistically independent configuration and momenta distributions:

$$P_R(\mathbf{R}) \propto \exp[-U(\mathbf{R})/k_BT] \tag{5}$$

$$P_P(\mathbf{P}) \propto \exp\left[-\sum_I \mathbf{P}_I^2/2M_ik_BT\right]. \tag{6}$$

As above, the momenta distribution factors into independent Gaussian distributions for each $\mathbf{P}_I$.

2.2. Relating atomistic and CG models

The previous subsection introduced atomistic and CG models without establishing any formal connection between them. We relate the two models by introducing a set of mapping functions. [80] We assume that each site physically represents a particular group of atoms in the atomistic model. We define a set of N linear mapping functions $\mathbf{M}(\mathbf{r}) = (\mathbf{M}_1(\mathbf{r}),\ldots,\mathbf{M}_N(\mathbf{r}))$ that define the Cartesian coordinates of each site as a linear function of atomic Cartesian coordinates with constant coefficients, c_{Ii}:

$$\mathbf{M}_I(\mathbf{r}) = \sum_i c_{Ii}\mathbf{r}_i, \tag{7}$$

for each site, $I = 1,\ldots,N$, in the CG model. If all of the atoms are translated by a vector displacement, then the mapping should translate each site by the same vector. Consequently, it follows that the coefficients, c_{Ii}, should satisfy the following normalization:

$$\sum_i c_{Ii} = 1, \tag{8}$$

for each site. Because we treat the sites as independent point particles, it follows that the CG momenta are simply $\mathbf{P}_I = M_I d\mathbf{R}_I/dt$. This implies a set of linear mapping functions $\mathbf{M}_\mathbf{P}(\mathbf{p}) = (\mathbf{M}_{\mathbf{P}1}(\mathbf{p}),\ldots,\mathbf{M}_{\mathbf{P}N}(\mathbf{p}))$ that define the Cartesian momenta of each site as a linear function of atomic Cartesian momenta:

$$\mathbf{M}_{\mathbf{P}I}(\mathbf{p}) = M_I\sum_i c_{Ii}\mathbf{p}_i/m_i, \tag{9}$$

for each site.

For each site, I, we define the set of involved atoms, $\mathcal{I}_\mathcal{I}$, as the set of atoms that provide a finite contribution to the sum in Eq. (7), i.e., the set for which $c_{Ii} \neq 0$. In principle, a single atom may be involved in multiple sites. In this case, it is also convenient to define a set of specific atoms that are involved in site I but are not involved in any other site. However, in the following we shall assume that, for each site, all of the atoms involved in the site are also specific to the site. (This last assumption can be readily relaxed. [80])

The above assumptions are sufficiently general that the present framework applies to a wide range of particle-based CG models. In particular, the present framework is valid when sites are placed at, e.g., the center-of-mass (i.e., $c_{Ii} \propto m_i$) or the center-of-geometry (i.e., $c_{Ii} = 1/n_I$, where n_I is the number of atoms involved in site I) of a group of atoms. This framework also applies for implicit solvent CG models [23, 63], in which entire molecules have been eliminated from the model and for mixed resolution models in which some sites represent single atoms, while others represent groups of atoms. [82] However, this framework does not apply for models, e.g., the adaptive resolution models [83] discussed in a later chapter by Delle Site and Praprotnik, for which molecules dynamically change resolution during the course of a simulation. [84] Nor does the following framework immediately apply for nonlinear coarse-graining maps.

2.3. Consistency criteria

The mapping operators defined by Eqs. (7)-(9), in combination with the atomistic probability distributions defined by Eqs. (2) and (3), imply probability distributions for sampling atomistic phase space points that map to a point in the phase space for the CG model. Because the atomistic configuration and momenta are statistically independent random variables and because the mapping operators do not mix configuration and momenta variables, the implied CG phase space distribution factors into statistically independent configuration and momentum distributions. In particular, the probability of sampling a particular atomistic configuration $\mathbf{r}$ that maps to a fixed CG configuration, $\mathbf{R}$, is given by

$$p_R(\mathbf{R}) = \langle \delta(\mathbf{M}(\mathbf{r}) - \mathbf{R}) \rangle \,, \tag{10}$$

where the angular brackets denote canonical ensemble averages over the atomistic model and $\delta(\mathbf{M}(\mathbf{r}) - \mathbf{R}) = \prod_I \delta(\mathbf{M}_I(\mathbf{r}) - \mathbf{R}_I)$. Similarly,

$$p_P(\mathbf{P}) = \langle \delta(\mathbf{M}_\mathbf{P}(\mathbf{p}) - \mathbf{P}) \rangle \tag{11}$$

is the probability of sampling a particular set of atomistic momenta $\mathbf{p}$ that map to a fixed set of CG momenta, $\mathbf{P}$. Because the CG model considers fewer particles than the original atomistic model, the mapping operators map multiple atomistic configurations to the same CG configuration. The distribution, $p_R(\mathbf{R})$ defined in Eq. (10) corresponds to the total Boltzmann weight of the volume element in the atomistic configuration space that maps to the specified point $\mathbf{R}$ in the CG configuration space. Similarly, $p_P(\mathbf{P})$ defined in Eq. (11) corresponds to the total Boltzmann weight of the volume element in the atomistic momentum space that maps to the specified point $\mathbf{P}$ in the CG momentum space.

A natural, though certainly not unique, definition for consistency with an atomistic model requires that a consistent CG model reproduce the distributions of CG phase variables that are implied by the atomistic probability distributions and CG mappings: [80]

$$P_R(\mathbf{R}) = p_R(\mathbf{R}) \tag{12}$$

$$P_P(\mathbf{P}) = p_P(\mathbf{P}). \tag{13}$$

These equations provide quantitative criterion for consistency between atomistic and CG models in configuration and momentum space, respectively.

The first criterion uniquely determines (to within an additive constant) the appropriate potential, U^0, for a CG model that is consistent with a particular atomistic model in configuration space:

$$U^0(\mathbf{R}) = -k_B T \ln p_R(\mathbf{R}) + \text{const} \tag{14}$$

$$= -k_B T \ln z(\mathbf{R}|u) + \text{const}, \tag{15}$$

in terms of a configuration-dependent partition function

$$z(\mathbf{R}|u) = \int d\mathbf{r}\ \exp[-u(\mathbf{r})/k_B T] \delta(\mathbf{M}(\mathbf{r}) - \mathbf{R}). \tag{16}$$

The notation $z(\mathbf{R}|u)$ in Eq. (15) implies that z is a function of the CG configuration $\mathbf{R}$ and a functional of the atomistic potential $u(\mathbf{r})$, i.e., it depends upon the value of $u(\mathbf{r})$ throughout the entire configuration space. [85] For reasons that will be emphasized later, U^0 is referred

to as a many-body potential of mean force (PMF). [19, 86] The many-body PMF is not a conventional potential because it considers both entropic and energetic contributions to give appropriate Boltzmann weight to each CG configuration. Consequently, the many-body PMF depends upon the system and thermodynamic state point. The constants in Eqs. (14) and (15) are independent of configuration and, therefore, do not impact the configuration probability distribution. However, they are dependent upon the thermodynamic state point. This state point dependence may be difficult to determine but can contain important information that impacts the thermodynamic properties of the CG model.

The previous analysis indicates that, for any atomistic model and any CG mapping, there exists a potential, U^0, that is sufficient to ensure that the CG model will be consistent with the underlying atomistic model in configuration space. [80] However, in general, it remains impractical to evaluate the integral of Eq. (16). Consequently, much of the remaining chapter will discuss various systematic approaches for approximating U^0. The development of CG potentials that not only accurately approximate the many-body PMF at a single thermodynamic state point for a single system, but are also transferable to multiple thermodynamic states and/or multiple systems remains a major challenge for the field.

While consistency in configuration space remains remarkably challenging, it is relatively easy to ensure consistency in momentum space between atomistic and CG models. A CG model will be consistent in momentum space with a particular atomistic model if and only if the following conditions are satisfied:[80]

1. Each atom is involved in at most one site.

2. The mass of each site is given by

$$M_I = \left(\sum_{i \in \mathcal{I}_I} \frac{c_{Ii}^2}{m_i} \right)^{-1} . \tag{17}$$

Note that this expression for the site mass depends upon both atomic masses and also mapping coefficients.

These conditions are obtained from the following considerations: Because each momentum variable $\mathbf{P}_I$ is an independent Gaussian random variable in the CG model, in a consistent CG model the distribution $p_P(\mathbf{P})$ in Eq. (11) must also factor into independent contributions for each $\mathbf{P}_I$. This is possible if and only if each atom is involved in at most one CG site.

The condition upon the CG site mass then follows because 1) the mapping expresses each CG momentum variable as the sum of atomistic momenta, each of which is an independent Gaussian random variable; and because 2) the sum of independent Gaussian random variables is a Gaussian random variable with a variance given by the sum of the variances for the contributing random variables. [87]

The consistency condition in momentum space has several implications. In the case that the mapping coefficients assign a site to the center of mass for an atomic group, i.e., $c_{Ii} \propto m_i$, then Eq. (17) implies that the appropriate mass for that site is the total mass of the atoms involved in the site. Moreover, for a CG model that is consistent in momentum space, the center-of-mass mapping maximizes the mass of a site relative to any other mapping that maps the same atoms to the site. In particular, if a group of atoms with unequal mass are involved in a site, then Eq. (17) leads to a smaller site mass for a center-of-geometry mapping than for a center-of-mass mapping.

According to our earlier definition in Eq. (13), a CG model that is consistent in momentum space will reproduce the distribution of fluctuations in momentum space. However, this definition does not imply or ensure any particular relationship between the dynamical trajectory observed during a CG MD simulation and any dynamical trajectory of the underlying atomistic model. Standard Mori-Zwanzig theory [88, 89] demonstrates that any conservative and deterministic atomistic trajectory will appear to be non-conservative and non-determininstic when observed in the CG phase space. In particular, the initial conditions in the phase space for a single CG trajectory correspond to an ensemble of initial conditions in the atomistic phase space. Moreover, dynamical simulations of CG models with conservative interaction potentials necessarily neglect fluctuating atomistic interactions that perturb and retard the atomic dynamics. Consequently, dynamical information obtained from CG MD simulations that sample the canonical ensemble for a conservative potential should necessarily be carefully considered. The time scale of CG dynamics is frequently estimated empirically by comparing the observed dynamics with those that are known from either detailed simulations or experiment. [90] However, the fundamental relationship between the dynamics of a CG model and the trajectories of an underlying atomistic model remains somewhat obscure. The development of CG models that accurately model the dynamics of complex fluids and molecular systems remains a topic of active and extensive research that is beyond the scope of the present review. See, e.g., the later chapter by Lu.

2.4. Maximum likelihood approach

Recently, Shell proposed a maximum likelihood framework that provides a complementary perspective for considering consistency between atomistic and CG models. [91] The maximum likelihood formalism assumes a mechanism for using the CG model to sample atomistic configurations. We shall use $P_r(\mathbf{r})$ to denote the probability that the CG model samples a configuration $\mathbf{R}$ that maps to an atomistic configuration $\mathbf{r}$. (We will consider this distribution in somewhat more detail momentarily.) The framework then considers the probability for the CG model to reproduce the distribution of configurations sampled by an atomistic model. In particular, if one discretizes the atomistic configuration space into distinguishable small volume elements of equal size and employs the CG model to perform n_s independent samples of the atomistic configuration space, then the probability that, during those n_s samples, each distinguishable configuration $\mathbf{r}$ is sampled $n(\mathbf{r})$ times is given by:

$$n_s! \prod_{\mathbf{r}} \frac{1}{n(\mathbf{r})!} P_r(\mathbf{r})^{n(\mathbf{r})}, \tag{18}$$

where $\mathbf{r}$ indexes the discrete cells in the configuration space and $W(\{n(\mathbf{r})\}) = n_s! / \prod_{\mathbf{r}} n(\mathbf{r})!$ is the multinomial factor corresponding to the number of distinguishable sequences of $n_s = \sum_{\mathbf{r}} n(\mathbf{r})$ configurations in which each distinguishable configuration $\mathbf{r}$ is represented $n(\mathbf{r})$ times. If the n_s configurations were sampled from the underlying atomistic model with a configuration probability distribution, $p_r(\mathbf{r})$, then the expected number of samples for the configuration $\mathbf{r}$ will be $n(\mathbf{r}) = n_s p_r(\mathbf{r})$. The probability that, during these n_s independent samples, the CG model samples each configuration $\mathbf{r}$ with the same frequency as the atomistic model does (i.e., a total of $n(\mathbf{r}) = n_s p_r(\mathbf{r})$ times in n_s samples) is given by the likelihood function:

$$L = n_s! \prod_{\mathbf{r}} \frac{1}{[n_s p_r(\mathbf{r})]!} P_r(\mathbf{r})^{[n_s p_r(\mathbf{r})]}. \tag{19}$$

The maximum likelihood approach parameterizes the CG model to maximize the probability of reproducing the distribution of atomistic configurations sampled by the atomistic model, i.e., the CG model should be parameterized to maximize the likelihood function L. [91]

Considering $n_s \to \infty$ samples, applying Stirling's approximation, [92] passing to a continuous description of the configuration space, and defining a dimensioned relative entropy

$S_{rel} = -(1/n_s)k_B \ln L$ leads to

$$S_{rel} = k_B \int d\mathbf{r}\; p_r(\mathbf{r}) \ln \left[\frac{p_r(\mathbf{r})}{P_r(\mathbf{r})} \right]. \tag{20}$$

The relative entropy corresponds to the Kullback-Leibler divergence from information theory [93] and quantitatively measures the overlap of the distributions of atomistic configurations generated by the atomistic and CG models, while weighting the overlap according to the atomistic distribution. Maximizing the likelihood function L corresponds to minimizing the relative entropy by definition. The Gibbs inequality, [85] which will be discussed in greater detail below, implies that $S_{rel} \geq 0$ and that $S_{rel} = 0$ if and only if $P_r(\mathbf{r}) = p_r(\mathbf{r})$ for all $\mathbf{r}$.

To proceed further in analyzing P_r we employ the identity:[87]

$$P_r(\mathbf{r}) = \int d\mathbf{R}\; P_{r|R}(\mathbf{r}|\mathbf{R}) P_R(\mathbf{R}), \tag{21}$$

where $P_R(\mathbf{R})$ is the probability distribution for the CG model to sample a configuration $\mathbf{R}$ and $P_{r|R}(\mathbf{r}|\mathbf{R})$ is the conditional probability of obtaining the atomistic configuration $\mathbf{r}$ given a fixed CG configuration $\mathbf{R}$. We assume that $P_{r|R}(\mathbf{r}|\mathbf{R}) = 0$ unless $\mathbf{R} = \mathbf{M}(\mathbf{r})$. Consequently, we consider the following form for $P_{r|R}$:

$$P_{r|R}(\mathbf{r}|\mathbf{R}) = \frac{g(\mathbf{r})\delta(\mathbf{M}(\mathbf{r}) - \mathbf{R})}{\Omega(\mathbf{R})}, \tag{22}$$

where $g(\mathbf{r})$ is a function of the atomistic configuration and $\Omega(\mathbf{R})$ is a normalization that quantifies the degeneracy of atomistic configurations that map to the fixed configuration:

$$\Omega(\mathbf{R}) = \int d\mathbf{r}\; g(\mathbf{r})\delta(\mathbf{M}(\mathbf{r}) - \mathbf{R}). \tag{23}$$

Two natural choices for $g(\mathbf{r})$ are 1) a uniform distribution, $g(\mathbf{r}) = 1$, [91] and 2) a Boltzmann weighted distribution, $g(\mathbf{r}) \propto \exp[-u(\mathbf{r})/k_B T]$, which was earlier considered by Krishna et al. in proposing a reweighting scheme for developing temperature transferable potentials. [94]

Employing Eqs. (21)-(23) in Eq. (20) and keeping track of dimensionless scaling, one obtains

$$S_{rel} = k_B \int d\mathbf{r}\; p_r(\mathbf{r}) \ln \left[\frac{V^n}{V^N} \frac{p_r(\mathbf{r})}{P_R(\mathbf{M}(\mathbf{r}))} \right] + S_{map} \tag{24}$$

where

$$S_{map} = k_B \left\langle \ln \left[\frac{V^N}{V^n} \frac{1}{g(\mathbf{r})} \Omega(\mathbf{M}(\mathbf{r})) \right] \right\rangle. \tag{25}$$

Equation (24) decomposes the relative entropy into two contributions: 1) a term which compares p_r and P_R and, therefore, explicitly depends upon the CG potential, U, according to Eq. (5); and 2) a term, S_{map}, which is independent of U but reflects the degeneracy of the CG mapping. Further analysis of S_{map} may provide a useful guide for designing $\mathbf{M}$.

A straightforward calculation employing Lagrange multipliers then demonstrates that the relative entropy is minimized when

$$P_R(\mathbf{R}) = p_R(\mathbf{R}), \tag{26}$$

which is the same consistency criterion described above. Consequently, the earlier consistency criterion not only corresponds to generating the same distribution of CG configurations as the underlying atomistic model, but also corresponds to minimizing the relative entropy between the two models, or equivalently, maximizing the overlap of the atomistic distributions generated by the two models. Recent work has expanded upon these considerations. [95, 96]

3. COMPUTATIONAL ALGORITHMS FOR CONSISTENCY

The previous section demonstrated that the many-body PMF defined in Eq. (14) is the appropriate potential for a CG model that is consistent with a particular atomistic model in configuration space, i.e., a model that reproduces the distribution of CG configurations generated by an atomistic model (and an associated mapping). However, in general, it is not feasible to calculate, represent, or simulate this many-body potential. Consequently, the development of accurate approximations to this many-body PMF remains a major challenge for systematic coarse-graining. This section presents and discusses some of the most popular and successful methods for determining CG potentials as a systematic approximation to the many-body PMF. We note that Larini and Krishna have recently proposed a generalized mean field theory for unifying the various CG methods discussed below (and also others). [97]

We will employ consistent notation in an attempt to unify the presentation and to clarify the relationships between different methods. Simultaneously, we will attempt to maintain sufficient generality that the presentation will apply to a wide class of CG models. In particular, we will focus on approximate CG potentials that are of the following form:

$$U(\mathbf{R}) = \sum_{\zeta} \sum_{\lambda} U_{\zeta}(\psi_{\zeta}(\{\mathbf{R}\}_{\lambda})), \tag{27}$$

where ζ indicates a particular interaction (e.g., a dihedral angle interaction) and U_ζ is the corresponding potential (e.g., a dihedral angle potential) that is a function of a single scalar variable, ψ_ζ, (e.g., a dihedral angle) that may be expressed as a function of the Cartesian coordinates, $\{\mathbf{R}\}_\lambda$, for a set of sites, λ (e.g., the 4 successively bonded sites that form a dihedral angle). This assumed form for the potential applies, e.g., for molecular mechanics potentials that are frequently employed in CG models. The following analysis can also be reformulated in the case that the approximate potential includes terms that depend upon more than a single variable, e.g., $U_\zeta(\psi_{\zeta 1}(\{\mathbf{R}\}_\lambda), \psi_{\zeta 2}(\{\mathbf{R}\}_\lambda))$, although that case will not be pursued here. Many CG methods bear striking similarities to classical density functional theories. [98] To emphasize this similarity, we shall also find it convenient to represent the approximate potential as a sum of terms, each of which is a product of a potential, U_ζ, and a conjugate field or density, $\hat{\rho}_\zeta$:

$$U(\mathbf{R}) = \sum_\zeta \int dx\; U_\zeta(x) \hat{\rho}_\zeta(\mathbf{R}; x), \tag{28}$$

where

$$\hat{\rho}_\zeta(\mathbf{R}; x) = \sum_\lambda \delta(\psi_{\zeta\lambda}(\mathbf{R}) - x) \tag{29}$$

in terms of $\psi_{\zeta\lambda}(\mathbf{R}) \equiv \psi_\zeta(\{\mathbf{R}\}_\lambda)$. The quantity $\hat{\rho}_\zeta$ defined by Eq. (29) is a density operator that can be interpreted as a counting device for determining, in a given configuration, $\mathbf{R}$, how many times the interaction potential $U_\zeta(x)$ contributes to the total potential. Ensemble averages of $\hat{\rho}_\zeta$ then determine structural correlation functions describing the interaction ζ.

Several methods determine approximate potentials that reproduce a set of lower order structural correlation functions, e.g., pair correlation functions, that are determined either from detailed simulations or experimental data. In particular, we shall consider atomistic (target) structural correlation functions of the form:

$$p_\zeta(x) = \langle \hat{\rho}_\zeta(\mathbf{R}; x) \rangle = \int d\mathbf{r}\; p_r(\mathbf{r}) \sum_\lambda \delta(\psi_{\zeta\lambda}(\mathbf{M}(\mathbf{r})) - x), \tag{30}$$

which describes the distribution of CG structures obtained by mapping configurations sampled from an atomistic model with a probability distribution $p_r(\mathbf{r})$. We shall also consider corresponding CG correlation functions:

$$P_\zeta(x) = P_\zeta(x|U) = \langle \hat{\rho}_\zeta(\mathbf{R}; x) \rangle_U = \int d\mathbf{R}\; P_R(\mathbf{R}|U) \sum_\lambda \delta(\psi_{\zeta\lambda}(\mathbf{R}) - x), \tag{31}$$

which describes the distribution of structures sampled from a CG model with probability distribution $P_R(\mathbf{R}|U)$. The subscripted angular brackets in Eq. (31) indicate a canonical ensemble average evaluated for a CG model with potential U, which is further emphasized in the notation $P_\zeta(x|U)$ and $P_R(\mathbf{R}|U)$. If ζ corresponds to the nonbonded interaction between a pair of site types in distinct molecules, then $p_\zeta(r)/r^2$ and $P_\zeta(R)/R^2$ are proportional to the corresponding site-site radial distribution functions (RDFs) generated from atomistic and CG simulations, respectively.

It should be emphasized that, even if a CG model reproduces a set of target lower order structural correlation functions, i.e., $\{P_\zeta(x) = p_\zeta(x) \text{ for all } \zeta\}$, it will not necessarily reproduce the higher order structural correlations generated by the atomistic model. [99] In particular, such a CG model is not guaranteed to reproduce the many-body distribution of sites generated by the atomistic model, i.e., $P_R(\mathbf{R}) \neq p_R(\mathbf{R})$. Furthermore, such a model is not guaranteed to reproduce any particular thermodynamic properties of the atomistic model.

3.1. Direct Boltzmann Inversion

If the interactions considered in the CG model are statistically independent of one another in the atomistic model, then p_R factors into independent terms for each interaction:

$$p_R(\mathbf{R}) \propto \prod_\zeta \prod_\lambda p_\zeta(\psi_{\zeta\lambda}(\mathbf{R})). \tag{32}$$

In this case, each term in the CG potential can be determined from "direct Boltzmann inversion" of the corresponding distribution:

$$U_\zeta(x) = -k_B T \ln[p_\zeta(x)/J_\zeta(x)], \tag{33}$$

where J_ζ corresponds to the Jacobian factor for the relevant order parameter, e.g., $J_\zeta(\theta) = \sin\theta$ for bond-angle interactions, $J_\zeta(\phi) = 1$ for dihedral angle interactions, and $J_\zeta(r) = r^2$ for central (radial) pair interactions. [100] CG potentials determined from direct Boltzmann inversion have enjoyed considerable success in polymeric systems, as reviewed, e.g., in Refs. (90) and (101). Direct Boltzmann inversion is often employed to parameterize bonded and other "hard" interactions that are only weakly coupled to the other interactions in the CG potential. [102] In this case, the remaining interaction potentials, e.g., nonbonded pair

potentials may be modeled with simple Lennard-Jones potentials to correctly reproduce excluded volume interactions or may be parameterized using iterative methods described below. [103, 104]

However, for some biological and soft matter systems, the interactions in the CG potential may be highly coupled and direct Boltzmann inversion may not quantitatively reproduce the target distribution functions. The importance of the correlations between the different potential terms and the corresponding difficulty that this can create for CG modeling is very clearly indicated by considering a CG model for a molecular fluid in which each molecule is represented by a single point site and the CG potential only includes central pair potentials between each pair of sites. In this case, $J_\zeta(r) \propto r^2$ so that $p_\zeta(r)/J_\zeta(r)$ is proportional to the RDF between molecules generated by the atomistic model, $g_\zeta(r)$, and the CG potential $U_\zeta(r)$ that is obtained from direct Boltzmann inversion is equivalent (to within an additive constant) to the pair potential of mean force (ppmf), $w_\zeta(r)$:

$$U_\zeta(r) = w_\zeta(r) = -k_B T \ln g_\zeta(r). \tag{34}$$

Because the RDF has maxima resulting from the packing of molecules into solvation shells, the ppmf also oscillates with attractive minima at the distances corresponding to the maxima in the RDF. The force function obtained from the ppmf, $-w'_\zeta(r)$, is equal to the magnitude of the average force experienced by a molecule averaged over all configurations in which a second molecule is a fixed distance r away. [85, 105] This force, which is referred to as the pair mean force, includes the average of the direct force between the pair, but also includes an environment mediated force, i.e., the average correlated force from the surrounding molecules. In general this contribution will not vanish; in fact, it is this contribution that generates the oscillations in the ppmf, which, in turn, result from the packing of the surrounding molecules into solvation shells.

If the ppmf is employed as a potential for simulating two sites in vacuum, then the environment mediated forces included in the ppmf will ensure that the distribution of distances sampled between the two sites in vacuum equals the distribution for the pair in condensed phase. However, if the ppmf is employed as a potential for simulating the molecules in the condensed phase, then these environment mediated interactions contributions to the ppmf will be included in the interaction potential between each pair of sites and, consequently, the CG model will systematically overcount these interactions. [106]

This problem is even more transparent in the trivial case in which both the atomistic and CG model correspond to the same simple Lennard-Jones fluid (i.e., no coarse-graining). In this case, the atomistic and CG pair potential should be the same, i.e., the original Lennard-Jones potential. [106] However, the CG potential determined by direct Boltzmann inversion (i.e., the ppmf) will oscillate with maxima at the peaks that are present in the atomistic RDF due to simple packing of spheres.

It is important to recognize the distinction between the ppmf, $w_\zeta(r)$, which describes the average force between a pair and which is determined by a two-body correlation function, and the many-body PMF, $U^0(\mathbf{R})$, which describes the average force on N sites and which is determined by an N-body distribution. [19, 86]

At present, there exists no direct (i.e., noniterative) method for determining CG potentials from structural correlation functions in such a way to ensure that the CG model will quantitatively reproduce a given set of correlation functions.

3.2. Iterative Boltzmann inversion

The iterative Boltzmann inversion (IBI) method is a natural extension of direct Boltzmann inversion that addresses the correlations between different interactions by iteratively modifying the CG potentials to quantitatively reproduce a given set of structural distribution functions. The IBI method was pioneered for CG models of fluids and polymers by Muller-Plathe and coworkers. [107] Schommers, [108] Soper, [109] and Reatto et al. [110] had earlier suggested somewhat similar approaches for determining empirical potentials for all-atom models of fluids. The IBI method also bears some similarity to the reverse Monte Carlo simulation method for iteratively determining structural ensembles from experimental scattering data. [111, 112]

The motivation for the IBI method can be easily understood by further analyzing the simple model described above, in which each molecule is represented by a single site and the CG potential only includes central pair potentials. [107, 109] Given a set of target site-site RDFs, $g_\zeta(r)$, obtained from either experiment or all-atom simulation, one can determine a set of approximate potentials $U_\zeta(r)$ from direct Boltzmann inversion. Simulations employing these potentials will then generate a set of CG site-site distribution functions, $G_\zeta(r)$. However, for the reasons described above, these CG RDFs may not quantitatively match the

target site-site RDFs, $g_\zeta(r)$. If, at some particular distance r_0, the RDF generated by the CG model is greater than the RDF generated by the atomistic model, i.e., $G_\zeta(r_0) > g_\zeta(r_0)$, this implies that the corresponding ppmf of the CG model, $W_\zeta(r) = -k_B T \ln G_\zeta(r)$, is more attractive than the ppmf of the atomistic model at the same distance, i.e., $W_\zeta(r_0) < w_\zeta(r_0)$. Consequently, the CG model will more accurately reproduce the atomistic RDF at r_0 if the CG ppmf is increased (made more repulsive) at that distance. Because the ppmf includes the direct interaction, the most straightforward way to increase the ppmf (and thus decrease the CG RDF) at r_0 is to simply increase the pair potential at that distance. A similar argument suggests that the CG RDF can be increased at r_0 by simply decreasing the pair potential at that distance. These ideas suggest that the CG potentials can be improved by modifying each term in the potential: [107, 109]

$$\delta U_\zeta(x) = \epsilon_\zeta (w_\zeta(x) - W_\zeta(x)) = -\epsilon_\zeta k_B T \ln \left[\frac{p_\zeta(x)}{P_\zeta(x|U)} \right], \tag{35}$$

where ϵ_ζ is a factor that may be adapted for different interactions or iterations. After modifying the potentials according to Eq. (35), the CG model is simulated again to generate a new set of structural correlation functions. These functions are compared with the target correlation functions; the CG potentials are modified according to Eq. (35); and, assuming convergence, the iterative procedure is repeated until the CG model reproduces the target distribution functions with sufficient accuracy.

The IBI method has been successfully used for a range of molecular liquids and polymeric systems. [113–121] The simplicity of the method is particularly appealing: each term in the potential is modified by considering the conjugate distribution function. However, the general convergence properties of the method are not very well understood. As described above, the ppmf includes both the direct interaction and also an environment mediated contribution. This latter contribution is only indirectly treated by the IBI method. In principle, because the different interactions may be highly coupled, modifying one particular potential can significantly impact the other distributions. Also, in some cases, the convergence of the IBI method has been sensitive to the order with which the different potentials are revised. [104, 122] In particular, it has been suggested that high frequency "hard" potentials should be optimized before low frequency "soft" potentials. [116, 123]

3.3. Inverse Monte Carlo

Lyubartsev and Laaksonen earlier introduced an inverse Monte Carlo (IMC) method for determining CG potentials that reproduce target distribution functions. [124, 125] The IMC method has been employed to quantitatively model various molecular systems, including liquids, [124] lipid bilayers, [126–128] and ionic systems. [129, 130] The IMC method has proven to be particularly amenable to theoretical characterization and several other related approaches have provided additional insight into the IMC approach.

Similarly to the IBI method, the IMC method employs iterative CG simulations to determine potentials $\{U_\zeta(x)\}$ that reproduce a set of target distributions, $\{p_\zeta(x)\}$. However, while the IBI method corrects each potential, U_ζ, independently according to Eq. (35), the IMC method employs linear response theory to estimate how variations in the potential U_ζ impact all of the distribution functions. A susceptibility matrix [124] describes how the distribution $P_\zeta(x|U)$ varies with small changes in $U_{\zeta'}(x)$.

$$M_{\zeta\zeta'}(x,x') \equiv \frac{\delta P_\zeta(x|U)}{\delta U_{\zeta'}(x')} = -\frac{1}{k_B T}\left\langle \Delta\hat{\rho}_\zeta(\mathbf{R};x)\Delta\hat{\rho}_{\zeta'}(\mathbf{R};x')\right\rangle_U , \tag{36}$$

where the second equality follows from employing Eq. (28) and differentiating Eq. (31), with

$$\Delta\hat{\rho}_\zeta(\mathbf{R};x) = \hat{\rho}_\zeta(\mathbf{R};x) - \left\langle\hat{\rho}_\zeta(\mathbf{R};x)\right\rangle_U . \tag{37}$$

The averages subscripted by U in Eqs. (36) and (37) are evaluated using the CG model with potential U according to Eq. (31), e.g., $\left\langle\hat{\rho}_\zeta(\mathbf{R};x)\right\rangle_U = P_\zeta(x|U)$. According to Eq. (36), the susceptibility matrix, $M_{\zeta\zeta'}(x,x')$, is proportional to the covariance matrix of fluctuations in the fields $\{\hat{\rho}_\zeta\}$ sampled by CG simulations. The linear analysis then relates variations in the CG potentials δU_ζ to variations in the CG distributions according to:

$$\delta P_\zeta(x) = \sum_{\zeta'}\int dx'\; M_{\zeta\zeta'}(x,x')\delta U_{\zeta'}(x'). \tag{38}$$

In the IMC method, $\delta P_\zeta(x)$ is defined by the difference between the target and CG simulated densities, $p_\zeta(x) - P_\zeta(x|U)$. Eq. (38) is then solved for $\delta U_{\zeta'}(x')$ to determine the corrections to the CG potential according to the Newton-Raphson method. [125] In practice, Eq. (38) is discretized and numerically solved as a system of coupled linear algebraic equations. [124] (Alternatively, as discussed further below, the potential may be expressed as a linear combination of discrete parameters which are refined using an analogous system

of linear equations.) The CG model is then simulated with the updated potential; the resulting distributions are compared with the target distributions; and the updated potentials are again corrected according to Eq. (38). Assuming convergence, the process is iterated until the final CG potentials reproduce the target distributions with sufficient accuracy.

3.4. Fundamental underpinnings of structure-focused CG methods

1. Useful tools

The Gibbs inequality [85] and generating functions [17] are useful tools for theoretical analyses of CG models.

1. **Gibbs inequality** Consider two positive functions (e.g., probability distributions), f and g that are defined on the same domain and have the same normalization, i.e.,

$$\int dx \, (f(x) - g(x)) = 0, \tag{39}$$

where x may be a high dimensional variable. It then follows that

$$\int dx \; f(x) \ln \left(\frac{f(x)}{g(x)} \right) \geq 0 \tag{40}$$

and the equality is only obtained when $f(x) = g(x)$ over the entire domain.

The Gibbs inequality implies the Gibbs-Bogoliubov-Feynman variational principle, [85] which is closely related to several variational principles for CG modeling. [49, 91] Within the canonical ensemble, the configuration probability distribution for a CG model with a potential U may be expressed:

$$P_R(\mathbf{R}|U) = V^{-N} \exp\left[-(U(\mathbf{R}) - F[U])/k_B T\right], \tag{41}$$

where $F[U]$ is the excess (nonideal) configurational contribution to the Helmholtz free energy for a CG model with potential U. If f and g are canonical configuration distributions associated with different potentials, $P_R(\mathbf{R}|U_A)$ and $P_R(\mathbf{R}|U_B)$, respectively, then Eq. (40) implies that:

$$F_B \leq F_A + \langle U_B(\mathbf{R}) - U_A(\mathbf{R}) \rangle_{U_A}, \tag{42}$$

where $F_A = F[U_A]$, $F_B = F[U_B]$, and the angular brackets denote a canonical ensemble average weighted by $P_R(\mathbf{R}|U_A)$.

The proof of the Gibbs inequality is straightforward. [85, 131] The left hand side of Eq. (40) can be expressed

$$\int dx\; g(x)\left\{z(x)\ln z(x) - z(x) + 1\right\}, \tag{43}$$

where $z(x) = f(x)/g(x)$, which is positive by hypothesis. The Gibbs inequality in Eq. (40) then immediately follows as a consequence of the fact that $z\ln z - z + 1 \geq 0$ is a convex function for $z > 0$ and achieves its minimum of 0 at $z = 1$.

2. **Generating functions** The configurational integral is a prototypical example of a generating function. [17]

$$Z[U] = V^N \exp[-F[U]/k_BT] \tag{44}$$

$$= \int d\mathbf{R} \exp\left[-\frac{1}{k_BT}\sum_\zeta \int dx\; U_\zeta(x)\hat{\rho}_\zeta(\mathbf{R};x)\right] \tag{45}$$

where the last line employs Eqs. (28) and (41). Differentiating $-F[U]/k_BT = \ln(V^{-N}Z[U])$ with respect to potentials, U_ζ, generates cumulants of the conjugate fields:

$$\frac{\delta}{\delta U_\zeta(x)}\ln(V^{-N}Z[U]) = -\frac{1}{k_BT}\left\langle\hat{\rho}_\zeta(\mathbf{R};x)\right\rangle_U \tag{46}$$

$$= -\frac{1}{k_BT}P_\zeta(x|U) \tag{47}$$

$$\frac{\delta^2}{\delta U_\zeta(x)\delta U_{\zeta'}(x')}\ln(V^{-N}Z[U]) = \left(\frac{1}{k_BT}\right)^2\left\langle\Delta\hat{\rho}_\zeta(\mathbf{R};x)\Delta\hat{\rho}_{\zeta'}(\mathbf{R};x')\right\rangle_U \tag{48}$$

$$= -\left(\frac{1}{k_BT}\right)M_{\zeta\zeta'}(x,x'). \tag{49}$$

Note that the second functional derivative is proportional to the susceptibility matrix considered by the IMC method.

2. Existence and Uniqueness

Given a set of target structural correlation functions, $\{p_\zeta(x)\}$, that correspond to a set of independent fields $\{\hat{\rho}_\zeta(\mathbf{R};x)\}$ in Eq. (30) and that are obtained from either atomistic

simulation or experimental data, two important questions arise: 1) Does there exist a set of CG interaction potentials, $\{U_\zeta(x)\}$, that will reproduce the target distribution functions? and 2) Assuming that such a set exists, is this set of potentials unique?

The answer to the second question is to the affirmative: At least in principle and assuming that they exist, the set of potentials that reproduces a set of distributions is unique. The proof of this assertion follows from a straightforward extension of the celebrated Henderson uniqueness theorem, [132] which is itself an extension of the Hohenberg-Kohn-Mermin theorem [133, 134] and follows from the Gibbs inequality. This result has been generalized for site-site intermolecular pair potentials. [135] We have recently further generalized the proof for more complex potentials. [96]

Consider two different CG potentials, U_A and U_B, that may be expressed in terms of two sets of potentials, $\{U_{\zeta A}(x)\}$ and $\{U_{\zeta B}(x)\}$, according to Eq. (28). We assume that the set of corresponding conjugate fields, $\{\hat{\rho}_\zeta(\mathbf{R};x)\}$, is linearly independent. These two potentials generate two sets of structural correlation functions $\{P_{\zeta A}(x) = P_\zeta(x|U_A)\}$ and $\{P_{\zeta B}(x) = P_\zeta(x|U_B)\}$. Because the fields are independent, it follows that, if the two sets of potentials differ (by more than a constant) in any one term, then $U_A(\mathbf{R}) - U_B(\mathbf{R})$ is not a constant. Consequently, the inequality is implied in the Gibbs inequality, Eq. (40). Employing the Gibbs-Bogoliubov-Feynman for both potentials according to Eq. (42) and combining the two relations leads to

$$\sum_\zeta \int dx\ \Delta U_\zeta(x) \Delta P_\zeta(x) < 0, \tag{50}$$

where $\Delta U_\zeta(x) = U_{\zeta A}(x) - U_{\zeta B}(x)$ and $\Delta P_\zeta(x) = P_{\zeta A}(x) - P_{\zeta B}(x)$. The inequality can only be achieved if $\Delta P_\zeta(x) \neq 0$ for some combination of ζ and x. Therefore, if two CG models generate the same set of distribution functions, $\{P_\zeta(x)\}$, then the two sets of interaction potentials, $\{U_\zeta(x)\}$, employed by the models are the same to within an additive constant.

The answer to the second question is much less straightforward. Chayes and coworkers provided a variational proof demonstrating that, for any $m-$body distribution function that is the reduction of an equilibrium distribution (for a possibly more complex potential), i.e., $\rho^{(m)}(\mathbf{R}_1, \ldots, \mathbf{R}_m) = \langle \delta(\mathbf{R}_1 - \mathbf{M}_1(\mathbf{r})) \cdots \delta(\mathbf{R}_m - \mathbf{M}_m(\mathbf{r})) \rangle$, there does indeed exist an $m-$body potential that will reproduce the $m-$body distribution. [136, 137] The proof is quite complicated and will not be further discussed here. Moreover, it is not clear to what extent this result can be generalized. In particular, it is not clear under what conditions

there exist molecular mechanics potentials of the form given by Eq. (28) that will reproduce a set of corresponding distribution functions.

3. Variational principles and properties

Several variational principles have been identified for determining CG potentials that reproduce a given set of target distribution functions. These approaches typically construct a functional of the CG potential, $\mathcal{F}[U]$, that achieves a stationary point when the target distribution functions are reproduced. Such a functional can be determined through two simple contributions:

1. The variational functional must couple each potential $U_\zeta(x)$ to the target atomistic distribution function, $p_\zeta(x)$. The simplest way to do so is to include a term of the form:
$$\sum_\zeta \int dx\; p_\zeta(x) U_\zeta(x) = \int d\mathbf{r}\; p_r(\mathbf{r}) U(\mathbf{M}(\mathbf{r})) = \int d\mathbf{R}\; p_R(\mathbf{R}) U(\mathbf{R}), \tag{51}$$
where we recall that p_R is the distribution of CG configurations implied by the atomistic distribution and CG mapping, so that Eq. (51) is the average of the CG potential weighted by the atomistic probability distribution.

2. Differentiation of the variational functional should also generate the corresponding CG distribution, $P_\zeta(x|U)$. The simplest way to do so is to include the generating function $F[U] = -k_B T \ln\left(V^{-N} Z[U]\right)$, according to Eq. (46).

By combining these terms, one obtains
$$\mathcal{F}[U] = \int d\mathbf{R}\; p_R(\mathbf{R}) \left(U(\mathbf{R}) - F[U]\right). \tag{52}$$
By construction,
$$\frac{\delta \mathcal{F}[U]}{\delta U_\zeta(x)} = p_\zeta(x) - P_\zeta(x|U), \tag{53}$$
so that $\mathcal{F}[U]$ achieves it stationary point when each target distribution is reproduced by the CG model. The quantity $\mathcal{F}[U]$ bears an intriguing similarity to an entropy function, $S = (U - F)/T$, for the CG model, but weights configurations according to the probability of sampling them from the atomistic model rather than from the CG model.

Murtola et al. [49] noted that $\mathcal{F}[U]$ is closely related to the functional employed in the density functional analysis of Chayes and coworkers [136, 137] and also to the relative entropy

S_{rel} considered by Shell. [91] By employing the canonical configuration distribution for the CG model, the functional $\mathcal{F}$ can also be expressed:

$$\mathcal{F}[U] = -k_B T \int d\mathbf{R}\; p_R(\mathbf{R}) \ln \left[V^N P_R(\mathbf{R}|U) \right] \tag{54}$$

$$= T \left(S_{rel}[U] + s[u] - S_{map} \right) \tag{55}$$

where $s[u]$ is the configurational entropy of the underlying atomistic model and S_{map} was defined in Eq. (25). Because $s[u]$ and S_{map} are both independent of U, it follows that the CG potential that minimizes the functional $\mathcal{F}[U]$ also minimizes the relative entropy, S_{rel}, and thus maximizes the overlap of the atomistic and CG configuration distributions.

Because the target distributions, $p_\zeta(x)$, are independent of the CG potential, Eq. (53) implies that the curvature of $\mathcal{F}[U]$, i.e., $\delta^2 \mathcal{F}[U]/\delta U_\zeta(x) \delta U_{\zeta'}(x')$, is proportional to the covariance matrix, $\langle \Delta\hat{\rho}_\zeta(\mathbf{R}; x) \Delta\hat{\rho}_{\zeta'}(\mathbf{R}; x') \rangle_U$, in Eq. (36) and is also proportional to the susceptibility matrix employed by the IMC method. Consequently, the IMC method corresponds to a Newton-Raphson method for finding the stationary point of $\mathcal{F}[U]$. [49, 125] Therefore, according to Eq. (55), the IMC method is also equivalent to the relative entropy approach for determining effective potentials. [49] This analysis also provides insight into the convergence properties of IMC and related approaches. [49] Because the covariance is everywhere positive semi-definite, it follows that $\mathcal{F}[u]$ has at most a single stationary point in the parameter space of potentials. However, these considerations are not sufficient to determine whether or not this stationary point exists.

Lyubartsev and Laaksonen also proved another intriguing variational property of the IMC method. [124] They demonstrated that the pair potentials that reproduce a given set of target pair correlation functions also maximizes the entropy of the CG model relative to any other model potential that reproduces the target correlation functions. This result also readily generalizes for molecular mechanics potentials: [96] Given a set of target structural correlation functions $\{p_\zeta(x)\}$ (and assuming such a potential exists), a molecular mechanics potential of the form given by Eq. (27) maximizes the entropy of the CG model relative to any other potential that reproduces the target correlation functions. This follows by applying the method of Lagrange multipliers to maximize the CG entropy with respect to variations in the CG distribution:

$$S[P_R] = -k_B \int d\mathbf{R}\; P_R(\mathbf{R}) \ln[V^N P_R(\mathbf{R})], \tag{56}$$

subject to the normalization condition:

$$\int d\mathbf{R}\ P_R(\mathbf{R}) - 1 = 0 \tag{57}$$

and the condition that each target correlation function is reproduced:

$$\int d\mathbf{R}\ P_R(\mathbf{R})\hat{\rho}_\zeta(\mathbf{R};x) - p_\zeta(x) = 0 \tag{58}$$

for each ζ and x.

Papoian and coworkers [138, 139] have provided further insight by emphasizing the coarse-graining procedure as a molecular renormalization group calculation and emphasizing the utility of finding a compact representation of the CG potential:

$$U(\mathbf{R}) = \sum_\alpha K_\alpha S_\alpha(\mathbf{R}), \tag{59}$$

where the (few) discrete potential parameters, K_α, are now interpreted as external fields, $U_\zeta(x) \rightarrow K_\alpha$. These fields then couple to (potentially many-body) structural observables, $\hat{\rho}_\zeta(\mathbf{R};x) \rightarrow S_\alpha(\mathbf{R})$, and determine a set of conjugate structural correlation functions, $P_\zeta(x|U) \rightarrow \langle S_\alpha(\mathbf{R})\rangle_U$. The CG potentials are optimized in a linear iterative renormalization scheme similar to the IMC approach in order to reproduce a corresponding set of target correlation functions. [124] The CG correlation functions correspond to derivatives of the partition function, $Z[U]$, with respect to the parameters, K_α, according to Eq. (46). Consequently, the IMC and relative entropy approaches can also be interpreted as renormalization-based methods for reproducing derivatives of the partition function.

3.5. Multiscale Coarse-graining

While the above methods parameterize potentials to reproduce lower order structural correlation functions, Izvekov and Voth have suggested a multiscale coarse-graining (MS-CG) method for determining CG potentials. [140–142] The MS-CG method employs a force-matching approach similar to that previously employed for ab initio simulations. [143, 144] The MS-CG method has been successfully applied for modeling a wide range of condensed phase systems, including molecular and ionic liquids, [106, 141, 145–147] lipid bilayers, [140, 148, 149] nanoparticles, [150] carbohydrates, [151] peptides, [152] and proteins. [153]

Rather than directly or iteratively addressing structural correlation functions, the MS-CG method employs force information sampled from atomistic simulations and employs this

information to directly approximate the many-body mean force field, [80, 106] i.e., the force field derived from gradients of the many-body PMF:

$$\mathbf{F}^0_I(\mathbf{R}) = -\frac{\partial U^0(\mathbf{R})}{\partial \mathbf{R}_I} = \langle \mathbf{f}_I(\mathbf{r}) \rangle_{\mathbf{R}} \,, \tag{60}$$

where the second relation follows from direct differentiation of Eq. (15) and

$$\mathbf{f}_I(\mathbf{r}) = \sum_{i \in \mathcal{I}_I} \mathbf{f}_i(\mathbf{r}) \tag{61}$$

is the net force on CG site I specified by the atomistic force field in configuration $\mathbf{r}$. The subscripted angular brackets in Eq. (60) denote a conditioned canonical ensemble average evaluated over all the atomistic configurations $\mathbf{r}$ that map to the fixed CG configuration $\mathbf{R}$:

$$\langle a(\mathbf{r}) \rangle_{\mathbf{R}} = \langle a(\mathbf{r}) \delta(\mathbf{M}(\mathbf{r}) - \mathbf{R}) \rangle \,/\, \langle \delta(\mathbf{M}(\mathbf{r}) - \mathbf{R}) \rangle \,, \tag{62}$$

for any quantity $a(\mathbf{r})$. Equation (60) demonstrates that U^0 is referred to as a "many-body potential of mean force" because the force field derived from its gradients is a many-body mean force field, i.e., a conditioned canonical ensemble average of the atomistic force $\mathbf{f}_I(\mathbf{r})$ experienced by the site I in the atomistic configurations, $\mathbf{r}$, that map to the fixed CG configuration, $\mathbf{R}$. [80, 86, 154] Because U^0 is determined from a many-body distribution function, the mean force $\mathbf{F}^0_I$ is generally a many-body function of the coordinates for all N sites. CG MD simulations that sample the canonical ensemble, while employing the many-body mean force field as a conservative force field, will generate a distribution of configurations that is consistent with the underlying atomistic model according to Eq. (12). [80]

To proceed in analyzing the MS-CG method it is convenient to consider a CG force field $\mathbf{F}$ as an object in an abstract vector space. [80, 155–157] Each element in this space specifies a set of N vector valued functions $(\mathbf{F}_1(\mathbf{R}), \ldots, \mathbf{F}_N(\mathbf{R}))$ that define a Cartesian force on each site I in any configuration $\mathbf{R}$. An inner product between two CG force fields $\mathbf{F}^{(1)}$ and $\mathbf{F}^{(2)}$ is defined:

$$\mathbf{F}^{(1)} \odot \mathbf{F}^{(2)} = \left\langle \sum_I \mathbf{F}^{(1)}_I(\mathbf{M}(\mathbf{r})) \cdot \mathbf{F}^{(2)}_I(\mathbf{M}(\mathbf{r})) \right\rangle . \tag{63}$$

It is important to note that this inner product corresponds to an ensemble average weighted by the atomistic equilibrium configuration distribution, $p_r(\mathbf{r})$. A norm is defined according to $||\mathbf{F}|| = (\mathbf{F} \odot \mathbf{F})^{1/2}$. Because the mean force field, $\mathbf{F}^0$, is a conditioned average of the atomistic force field, $\mathbf{f}$, the following identity holds for any CG force field, $\mathbf{F}$: [80, 155]

$$\left\langle \sum_I \mathbf{f}_I(\mathbf{r}) \cdot \mathbf{F}_I(\mathbf{M}(\mathbf{r})) \right\rangle = \mathbf{F}^0 \odot \mathbf{F}. \tag{64}$$

The MS-CG force field is determined by minimizing a force-matching [143, 144] functional defined on the space of CG force fields: [80, 140]

$$\chi^2[\mathbf{F}] = \frac{1}{3N} \left\langle \sum_I |\mathbf{f}_I(\mathbf{r}) - \mathbf{F}_I(\mathbf{M}(\mathbf{r}))|^2 \right\rangle \tag{65}$$

As a direct consequence of Eq. (64), the MS-CG functional may be re-expressed:

$$\chi^2[\mathbf{F}] = \chi^2[\mathbf{F}^0] + \frac{1}{3N}||\mathbf{F}^0 - \mathbf{F}||^2, \tag{66}$$

where $\mathbf{F}^0$ is the many-body mean force field. The second term $||\mathbf{F} - \mathbf{F}^0||^2 \geq 0$ for all CG force fields, $\mathbf{F}$, and $||\mathbf{F} - \mathbf{F}^0||^2 = 0$ if and only if $\mathbf{F} = \mathbf{F}^0$, i.e., if and only if $\mathbf{F}_I(\mathbf{R}) = \mathbf{F}_I^0(\mathbf{R})$ for all I and $\mathbf{R}$. Consequently, the CG force field that provides the global minimum of χ^2 is the many-body mean force field and the resulting CG model will be consistent with the underlying atomistic model from which the forces $\mathbf{f}_I$ were sampled. Intriguingly, recent work has demonstrated that χ^2 can also be expressed as the gradient squared of the information function employed in the relative entropy approach. [96]

Equations (65) and (66) also have an intuitive geometric interpretation. The vector space of CG force fields described above may be considered a subspace of a larger vector space of force fields that depend upon the atomistic configuration. The atomistic force field $\mathbf{f} = (\mathbf{f}_1(\mathbf{r}), \ldots, \mathbf{f}_N(\mathbf{r}))$ defined in Eq. (61) is a particular element in this larger vector space. The functional $\chi^2[\mathbf{F}]$ then corresponds to the "distance" of a particular CG force field $\mathbf{F}$ from the atomistic force field, $\mathbf{f}$. According to Eq. (60) the mean force field $\mathbf{F}^0$ is a conditioned average of the atomistic force field, which corresponds to a geometric projection from the larger vector space into the vector subspace of CG force fields. [88, 89, 154–159] Equivalently, Eq. (64) implies that the vector $\mathbf{f} - \mathbf{F}^0$ is orthogonal to any vector in the subspace of CG force fields. [155] By geometric arguments, the projection of a vector (i.e., $\mathbf{f}$) into a subspace determines the element (i.e., $\mathbf{F}^0$) that is minimum distance from the original vector. Thus, Eq. (66) corresponds to a simple statement of the Pythagorean theorem within the space of CG force fields: The square of the distance of the atomistic force field, $\mathbf{f}$, from a trial CG force field, $\mathbf{F}$, is equal to the sum of two terms that correspond to the orthogonal sides of a right triangle: 1) the square of the distance from the atomistic force field to the mean force field, i.e., $\chi^2[\mathbf{F}^0]$; and 2) the square of the distance from the mean force field to the trial CG force, i.e., $||\mathbf{F} - \mathbf{F}^0||^2$. Because the first component is fixed by the original atomistic model and the CG mapping, minimizing the distance of a trial CG force field, $\mathbf{F}$, from the

atomistic force field, $\mathbf{f}$, (i.e., minimizing $\chi^2[\mathbf{F}]$) corresponds to minimizing the distance of the trial force field from the mean force field (i.e., minimizing $||\mathbf{F} - \mathbf{F}^0||^2$). This distance reaches its global and unique minimum when the two force fields are equal.

As noted above for the many-body PMF, it is generally impossible to calculate, represent, or simulate the many body mean force field, $\mathbf{F}^0$. In the context of the MS-CG theory, doing so would correspond to minimizing χ^2 with a complete basis set for the space of CG force fields, which is completely intractable for any complex molecular system. Consequently, it is important to identify practically useful incomplete basis sets for CG force fields. [155] The following analysis will consider the basis set relevant for molecular mechanics potentials of the form given by Eq. (27), although more complex potentials have also been used. [160] Such approximate potentials are frequently employed and often provide a reasonably accurate approximation to the many-body mean force field. However, it is important to recognize that such a basis set is highly incomplete. At this time it remains unclear under which circumstances such an incomplete basis set will allow an accurate approximation to the PMF.

We represent each term in the approximate potential with a discrete set of basis functions of a single variable, $U_{\zeta d}(x)$, with constant coefficients $\phi_{\zeta d}$ that will serve as force field parameters: [80, 155]

$$U_\zeta(x) = \sum_d \phi_{\zeta d} U_{\zeta d}(x). \tag{67}$$

A wide range of basis functions may be usefully used. [96, 155, 157, 161–163] For instance, polynomial functions have been employed for harmonic and anharmonic bonded potentials; cosine series have been employed for torsional potentials; and discrete delta functions ("hat functions") or flexible spline functions have been employed in representing bonded potentials and also nonbonded pair potentials. This basis expansion of the potentials determine a basis expansion of the corresponding force functions:

$$F_\zeta(x) = \sum_d \phi_{\zeta d} f_{\zeta d}(x), \tag{68}$$

where $F_\zeta(x) = -dU_\zeta(x)/dx$ and $f_{\zeta d}(x) = -dU_{\zeta d}(x)/dx$. The force on site I may be expressed:

$$\mathbf{F}_I(\mathbf{R}) = \sum_\zeta \sum_d \phi_{\zeta d}\, \boldsymbol{\mathcal{G}}_{I;\zeta d}(\mathbf{R}), \tag{69}$$

where

$$\boldsymbol{\mathcal{G}}_{I;\zeta d}(\mathbf{R}) = \sum_{\lambda} \frac{\partial \psi_{\zeta\lambda}(\mathbf{R})}{\partial \mathbf{R}_I} f_{\zeta d}(\psi_{\zeta\lambda}(\mathbf{R})). \tag{70}$$

By using the force field representation introduced above and replacing the sums over ζ and d with a single sum over a "super-index" $D = \zeta d$, Eq. (69) may be re-expressed: [155]

$$\mathbf{F}(\phi) = \sum_{D} \phi_D \boldsymbol{\mathcal{G}}_D. \tag{71}$$

In this representation, the CG force field, $\mathbf{F}$, has been expressed as a linear combination of force field vectors, $\boldsymbol{\mathcal{G}}_D$, the components of which are given by Eq. (70). The set of vectors included in Eq. (71) define an incomplete basis set that spans a vector subspace of CG force fields. The force field parameters $\{\phi_D\}$ serve as constant coefficients that identify a particular force field within this subspace. [80]

Employing this basis set expansion for the force field, the MS-CG functional becomes a simple quadratic function of the force field parameters:

$$\chi^2(\phi) = \chi^2[\mathbf{F}(\phi)] = \frac{1}{3N} \left\langle \sum_{I} \left| \mathbf{f}_I(\mathbf{r}) - \sum_{D} \phi_D \boldsymbol{\mathcal{G}}_{I;D}(\mathbf{M}(\mathbf{r})) \right|^2 \right\rangle. \tag{72}$$

The same geometric arguments that determined the many-body mean force field as a projection of the atomistic force field now apply for the vector subspace spanned by a given basis set. The function $\chi^2(\phi)$ may be re-expressed:

$$\chi^2(\phi) = \chi^2[\mathbf{F}^0] + \frac{1}{3N}||\mathbf{F}^0 - \mathbf{F}(\phi)||^2 \tag{73}$$

$$= \chi^2[\mathbf{F}^0] + \frac{1}{3N}||\mathbf{F}^0 - \mathbf{F}(\phi^*)||^2 + \frac{1}{3N}||\mathbf{F}(\phi^*) - \mathbf{F}(\phi)||^2, \tag{74}$$

where $\mathbf{F}(\phi^*) = \sum_D \phi_D^* \boldsymbol{\mathcal{G}}_D$ corresponds to the geometric projection of the mean force $\mathbf{F}^0$ into the subspace spanned by the basis set $\{\boldsymbol{\mathcal{G}}_D\}$. This force field then provides the unique global minimum of $\chi^2(\phi)$ relative to any force field within this subspace (i.e., relative to any force field of the form given by Eq. (69)). By minimizing $\chi^2(\phi)$, $\mathbf{F}(\phi^*)$ also minimizes the distance $||\mathbf{F}^0 - \mathbf{F}(\phi)||^2$ and is, therefore, the force field within the subspace that is "closest" to the many-body mean force. Moreover, this force field is unique for a given basis set, although, if the basis vectors are not linearly independent, then $\mathbf{F}(\phi^*)$ does not determine a unique set of parameters ϕ^*. [155] The present framework also applies in the case that a fixed "reference" contribution from, e.g., electrostatic interactions, is included in the CG potential. [155]

This analysis forms the basis of the MS-CG variational principle for determining an optimized CG potential using forces sampled from atomistic MD simulations. [80, 140, 141, 155, 163] If one minimizes $\chi^2[\mathbf{F}]$ while using a complete force field basis set, then the resulting force field will be the many-body mean force field [106] and the associated CG model will be completely consistent with the underlying atomistic model. [80] If one minimizes $\chi^2(\phi)$ while using an incomplete force field basis set, then the resulting CG force field will be the projection of the mean force field onto the given basis. [155] This force field will then be an "optimal approximation" to the mean force field in the sense that it minimizes the distance from the mean force field, $||\mathbf{F}^0 - \mathbf{F}(\phi)||^2$, relative to any other force field spanned by the basis. [80, 159]

This corresponds to a different variational principle (or different metric) than is considered by the iterative structure-motivated CG methods. In general, given the same incomplete basis set, the MS-CG potential will differ from the potentials that are obtained from iterative structure-motivated CG methods. Iterative structure-motivated methods (e.g., IBI or IMC) are designed to determine the CG potential that reproduces a set of target distribution functions. However, CG MD simulations that sample the canonical ensemble using the MS-CG potential are not guaranteed to quantitatively reproduce any particular distribution function. Rather, in the case of the MS-CG method, distribution functions are predictions of the method. [106]

Importantly, Eq. (72) demonstrates that the MS-CG potential is determined from a linear least squares problem, for which powerful numerical techniques have been developed. [164, 165] Two numerical approaches have been developed for determining the MS-CG potential. [155, 163] In principle, both methods obtain identical potentials. Both methods employ a sequence of N_t sets of atomistic configurations and forces that have been sampled by either MD or MC simulation, $\{\mathbf{r}_t, \mathbf{f}(\mathbf{r}_t)\}$. The function $\chi^2(\phi)$ may then be expressed:

$$\chi^2(\phi) = \frac{1}{3N_tN} \sum_t \sum_I \left| \mathbf{f}_I(\mathbf{r}_t) - \sum_D \phi_D \boldsymbol{\mathcal{G}}_{I;D}(\mathbf{M}(\mathbf{r}_t)) \right|^2 , \tag{75}$$

which may then be minimized directly by methods such as QR decomposition or singular value decomposition. If the calculation involves too many sites, parameters, and configurations for the minimization to be performed in memory, then χ^2 can be approximately solved via a block-averaging approach in which configurations are partitioned into smaller blocks, and χ^2 is expressed as a sum of residuals corresponding to each block. The optimal set of

parameters are then approximated by averaging the parameters that minimize χ^2 for each block. [141] In practice, the block-averaging approximation has provided an accurate and stable numerical method for determining the parameters for the MS-CG potential for a wide range of systems. [163]

Alternatively, the sampled configurations and forces can be used to approximate equilibrium correlation functions and χ^2 can be expressed:

$$\chi^2(\phi) = \chi^2(0) - \frac{2}{3N}\sum_D b_D\phi_D + \frac{1}{3N}\sum_D\sum_{D'}\phi_D G_{DD'}\phi_{D'}, \tag{76}$$

where

$$b_D = \left\langle \sum_I \mathbf{f}_I(\mathbf{r}) \cdot \boldsymbol{\mathcal{G}}_{I;D}(\mathbf{M}(\mathbf{r})) \right\rangle \tag{77}$$

$$G_{DD'} = \left\langle \sum_I \boldsymbol{\mathcal{G}}_{I;D}(\mathbf{M}(\mathbf{r})) \cdot \boldsymbol{\mathcal{G}}_{I;D'}(\mathbf{M}(\mathbf{r})) \right\rangle. \tag{78}$$

Minimizing Eq. (76) with respect to ϕ_D leads to the normal system of equations: [106, 155, 163]

$$\sum_{D'} G_{DD'}\phi_{D'} = b_D. \tag{79}$$

The system of normal equations have the advantage that they require substantially less computational memory. However, the condition number of the matrix $G_{DD'}$ is the square of the condition number of the matrices involved in directly minimizing χ^2 using the approaches described in the previous paragraph. Nevertheless, the normal equations have also been successfully applied for a variety of different systems. [106, 155–157, 162]

The normal equations formulated in Eqs. (77) further emphasize the geometric interpretation of the MS-CG method. [156, 157] From the identity of Eq. (64), it follows that b_D corresponds to the inner product of the mean force field, $\mathbf{F}^0$, with the basis vector $\boldsymbol{\mathcal{G}}_D$:

$$b_D = \mathbf{F}^0 \odot \boldsymbol{\mathcal{G}}_D. \tag{80}$$

Similarly, $G_{DD'}$ is a metric tensor that corresponds to the inner product between basis vectors: $G_{DD'} = \boldsymbol{\mathcal{G}}_D \odot \boldsymbol{\mathcal{G}}_{D'}$. The normal equations determine the MS-CG force field, $\mathbf{F}$, as a projection of the mean force field, $\mathbf{F}^0$, into the subspace spanned by $\{\boldsymbol{\mathcal{G}}_D\}$ by requiring that the MS-CG and mean force field have the same inner product with each basis vector:

$$b_D = \boldsymbol{\mathcal{G}}_D \odot \mathbf{F}^0 = \boldsymbol{\mathcal{G}}_D \odot \mathbf{F} \tag{81}$$

which is equivalent to determining the vector $\mathbf{F}$ that makes $\mathbf{F}^0 - \mathbf{F}$ orthogonal to every force field spanned by the basis. Because the inner product in the space of force fields corresponds to an ensemble average and because the different basis vectors correspond to molecular interactions that are correlated in a condensed phase, the basis vectors are not orthogonal and $G_{DD'}$ is not diagonal. Rather, the off-diagonal elements of $G_{DD'}$ quantify the correlations between the different interactions included in the potential. [156, 157] If the different terms in the potential were statistically independent, then the metric tensor $G_{DD'}$ would be diagonal and each force field parameter, ϕ_D, could be directly determined as b_D/G_{DD}. [106] This corresponds to the assumption that is implicit in direct Boltzmann inversion. However, in contrast to the IBI or IMC methods, which neglect these many-body correlations in determining a first trial potential and then address the correlations by iteratively correcting the potential, the MS-CG method employs the metric tensor to directly disentangle these correlations. These considerations are expanded in the following subsection.

3.6. generalized-YBG theory

The above discussion naturally emphasized the importance of atomistic forces in determining the MS-CG potential. Therefore, it is somewhat surprising that the MS-CG potential can also be directly determined from structural correlation functions. [106, 156] The resulting analysis further clarifies the connection between the MS-CG method and other structure-motivated methods. This correspondence is most conveniently explored by employing a continuous basis set to represent CG force fields. [157] In this case, we consider potentials of the form given by Eq. (28) and the corresponding force field may be expressed:

$$\mathbf{F} = \sum_{\zeta} \int dx \, \phi_\zeta(x) \, \boldsymbol{\mathcal{G}}_\zeta(x), \tag{82}$$

where $\phi_\zeta(x) = -dU_\zeta(x)/dx$ is a force function and $\boldsymbol{\mathcal{G}}_\zeta(x)$ is a corresponding force field vector with elements

$$\boldsymbol{\mathcal{G}}_{I;\zeta}(\mathbf{R}; x) = \sum_{\lambda} \frac{\partial \psi_{\zeta\lambda}(\mathbf{R})}{\partial \mathbf{R}_I} \delta(\psi_{\zeta\lambda}(\mathbf{R}) - x). \tag{83}$$

We note in passing that the present argument can be generalized to the case that the CG potential includes a fixed reference term. [166]

When expressed in the continuous basis set, the MS-CG functional becomes [157]

$$\chi^2[\{\phi_\zeta(x)\}] = \chi^2[0] - \frac{2}{3N}\sum_\zeta \int dx\; b_\zeta(x)\phi_\zeta(x) \tag{84}$$

$$+\frac{1}{3N}\sum_\zeta\sum_{\zeta'}\int dx \int dx' \phi_\zeta(x) G_{\zeta\zeta'}(x,x')\phi_{\zeta'}(x'), \tag{85}$$

and the normal equations may be expressed:

$$\sum_{\zeta'}\int dx' G_{\zeta\zeta'}(x,x')\phi_{\zeta'}(x') = b_\zeta(x). \tag{86}$$

in terms of

$$b_\zeta(x) = \boldsymbol{\mathcal{G}}_\zeta(x) \odot \mathbf{F}^0 = \left\langle \sum_I \mathbf{f}_I(\mathbf{r}) \cdot \boldsymbol{\mathcal{G}}_{I;\zeta}(\mathbf{M}(\mathbf{r});x) \right\rangle \tag{87}$$

$$G_{\zeta\zeta'}(x,x') = \boldsymbol{\mathcal{G}}_\zeta(x) \odot \boldsymbol{\mathcal{G}}_{\zeta'}(x') = \left\langle \sum_I \boldsymbol{\mathcal{G}}_{I;\zeta}(\mathbf{M}(\mathbf{r});x) \cdot \boldsymbol{\mathcal{G}}_{I;\zeta'}(\mathbf{M}(\mathbf{r});x') \right\rangle, \tag{88}$$

where $b_\zeta(x)$ is the inner product of the many-body mean force field with the basis vector $\boldsymbol{\mathcal{G}}_\zeta(x)$ and $G_{\zeta\zeta'}(x,x')$ is a many-body structural correlation function that corresponds to a metric tensor defined by the given basis set. The metric tensor $G_{\zeta\zeta'}(x,x')$ should be compared with the susceptibility matrix, $M_{\zeta\zeta'}(x,x')$, employed in IMC and defined in Eq. (36). While $M_{\zeta\zeta'}(x,x')$ considers the correlations between every set of interactions in the CG potential, $G_{\zeta\zeta'}(x,x')$ trivially vanishes for any two potentials that do not act on a common site.

Equation (87) indicates that the MS-CG method employs atomistic force information as a surrogate for information regarding the many-body mean force field. [156, 157] Consequently, $b_\zeta(x)$ can be re-expressed:

$$b_\zeta(x) = \int d\mathbf{R}\; p_R(\mathbf{R}) \sum_I \boldsymbol{\mathcal{G}}_{I;\zeta}(\mathbf{R};x) \cdot \mathbf{F}_I^0(\mathbf{R}) \tag{89}$$

$$= k_B T \int d\mathbf{R} \sum_I \boldsymbol{\mathcal{G}}_{I;\zeta}(\mathbf{R};x) \cdot \frac{\partial}{\partial \mathbf{R}_I} p_R(\mathbf{R}) \tag{90}$$

where the second relation expresses the many-body mean force as gradients of the many-body PMF which is related to p_R according to Eq. (14). Performing integration by parts, $b_\zeta(x)$ can be expressed entirely in terms of structural correlation functions: [156, 157]

$$b_\zeta(x) = k_B T \left(\frac{d}{dx}\bar{g}_\zeta(x) - L_\zeta(x) \right). \tag{91}$$

One then arrives at a system of linear integral equations for determining the MS-CG potential from structural correlation functions, $G_{\zeta\zeta'}(x,x')$, $\bar{g}_\zeta(x)$, and $L_\zeta(x)$:

$$\sum_{\zeta'} \int dx' G_{\zeta\zeta'}(x,x')\phi_{\zeta'}(x') = k_B T \left(\frac{d}{dx}\bar{g}_\zeta(x) - L_\zeta(x) \right) \tag{92}$$

where

$$\bar{g}_\zeta(x) = \left\langle \sum_\lambda \left|\nabla\psi_{\zeta\lambda}(\mathbf{M}(\mathbf{r}))\right|^2 \delta(\psi_{\zeta\lambda}(\mathbf{M}(\mathbf{r})) - x) \right\rangle \tag{93}$$

$$L_\zeta(x) = \left\langle \sum_\lambda \nabla^2\psi_{\zeta\lambda}(\mathbf{M}(\mathbf{r}))\delta(\psi_{\zeta\lambda}(\mathbf{M}(\mathbf{r})) - x) \right\rangle, \tag{94}$$

$|\nabla\psi_{\zeta\lambda}(\mathbf{R})|^2 = \sum_I (\partial\psi_{\zeta\lambda}(\mathbf{R})/\partial\mathbf{R}_I)^2$, and $\nabla^2\psi_{\zeta\lambda}(\mathbf{R}) = \sum_I \partial^2\psi_{\zeta\lambda}(\mathbf{R})/\partial\mathbf{R}_I^2$.

This system of linear integral equations, Eq. (92), is a generalization for molecular systems [156, 157] of the Yvon-Born-Green theory, which is a well known integral equation theory for simple liquids. [85, 105] If, for instance, the CG potential is composed of simple pair potentials, $U(\mathbf{R}) = \sum_\lambda U^{(2)}(\psi_\lambda(\mathbf{R}))$, where λ identifies a particular pair of sites $\{I,J\}$, $\psi_\lambda(\mathbf{R}) = |\mathbf{R}_I - \mathbf{R}_J|$ is the distance between the pair, and $U^{(2)}(r)$ is a central pair potential. Then $|\nabla\psi_\lambda(\mathbf{R})|^2 = 2$, $\nabla^2\psi_\lambda(\mathbf{R}) = 4/\psi_\lambda(\mathbf{R})$, $\bar{g}(r) = 2\left\langle \sum_\lambda \delta(\psi_\lambda(\mathbf{R}) - r)\right\rangle$, and $L(r) = 2\bar{g}(r)/r$. In this case, $G_{\zeta\zeta'}(r,r')$ is a spherically averaged three-body correlation function and $\bar{g}(r)$ is related to the RDF, $g(r) = \exp[-w(r)/k_BT]$, according to $\bar{g}(r) = 2r^2 g(r)/c$, where c is a dimensioned normalization. It also follows that $b(r)$ is closely related to the pair mean force, $-w'(r)$:

$$b(r) = (2r^2/c)k_BTg'(r) = (2r^2/c)[-w'(r)g(r)]. \tag{95}$$

After dividing through by the RDF, $g(r)$, the generalized YBG equation Eq. (92) may be re-expressed:

$$-w'(r) = \phi(r) + \int dr' \left(\frac{c}{2r^2}\right) g^{-1}(r)\bar{G}(r,r')\phi(r'), \tag{96}$$

where $\bar{G}(r,r')$ is the explicitly many-body contribution to the metric tensor $G(r,r') = \bar{g}(r)\delta(r-r') + \bar{G}(r,r')$. Equation (96) is an explicit restatement of the conventional YBG theory that states that the pair mean force, $-w'(r)$, which (as discussed earlier) is the average force on a pair of particles separated by a distance r, can be decomposed into two contributions: 1) the direct force between the pair at a fixed distance r, $\phi(r) = -dU^{(2)}(r)/dr$; and 2) the average correlated force from the environment on a pair of particles separated by r. The

integral term in Eq. (96) decomposes this environment-mediated force into contributions from shells of radius r' from the fixed particles.

The YBG theory of simple liquids had previously been considered to be an exact direct relation between a pair potential and the equilibrium two and three-particle correlation functions generated by the pair potential. [85, 105] This direct relationship had also been extended for rigid anisotropic molecules [167] and for polymers modeled with central pair potentials and/or rigid bonds. [168–172] More recently, Weeks and coworkers have exploited the YBG theory to model long-ranged electrostatics. [173–175] However, the theory derived above in Eq. (92) generalizes the YBG theory for arbitrarily complex molecules with flexible internal degrees of freedom modeled with molecular mechanics potentials. [156] Moreover, the above discussion demonstrates that the generalized-YBG theory is also a powerful method for determining an optimal approximation of a given form to an unknown and arbitrarily complex potential directly (i.e., noniteratively) from structural correlation functions. [156] In the case of CG modeling, the arbitrarily complex potential is the many-body PMF, the structural correlation functions reflect distributions of CG sites, the form of the approximate potential is determined from the force field basis set implicit in Eq. (92), and the optimal approximate potential corresponds to the MS-CG potential. Consequently, the g-YBG theory determines the MS-CG potential directly from structural correlation functions, [106, 157] which can be obtained, perhaps, from a protein databank. [176]

This analysis further clarifies the MS-CG method as a force-balance relation between the atomistic and CG model. The CG potentials, $U_\zeta(x)$, are determined to balance the average atomistic force along each basis vector. Higher-order structural correlations clearly play an important role in redistributing the many-body mean force among the various potentials. However, even if a CG model reproduces a set of lower order structural correlation functions, in general, the model will not accurately reproduce the higher order many-body structural correlation functions of an atomistic model. [99] Consequently, the MS-CG potential may not exactly reproduce the atomistic average force or, equivalently, the atomistic distribution for each degree of freedom. [106]

There are, in fact, two relevant g-YBG equations when considering CG potentials: [49, 106]

1. The set of structural correlation functions generated by the atomistic model determines one g-YBG equation. The MS-CG potential satisfies this (inverse) g-YBG equation

and is the projection of the many-body mean force field onto an incomplete basis set.

2. The set of structural correlation functions generated by a given CG potential determines a second g-YBG equation. Even if the CG potential reproduces the lower order structural correlation functions generated by the atomistic model, the higher order structural correlations will be different. Consequently, the second g-YBG equation is distinct from the first g-YBG equation. The CG potential will be an exact (direct) solution of this second g-YBG equation.

Chu and coworkers have exploited this logic in developing an iterative-YBG approach that systematically corrects the potential so that, for each relevant degree of freedom, the direct and indirect forces generated by the CG model balance the corresponding mean force generated by the atomistic model. [177]

4. OUTSTANDING CHALLENGES

The promised efficiency of CG models has attracted tremendous recent interest. Consequently, dramatic progress has been rapidly made in the development of computational algorithms and theoretical understanding for systematic coarse-graining methods that ensure consistency with high resolution models. Nevertheless, systematic CG simulation methods are still in their infancy in comparison to the well established atomistic simulation methods that have been intensely studied and optimized for decades. Extensive computational and theoretical study is still required before CG models might attain the accuracy, transferability, and predictive capability of all-atom models. The present section briefly discusses a few of the most significant outstanding challenges facing the field of systematic coarse-graining.

4.1. Transferability

One of the most important and well known challenges facing CG models regards their transferability. The present chapter focused on methods that attempt to systematically determine accurate approximations to the many-body PMF. Equation (16) demonstrates that the PMF depends upon the system composition and volume by defining the configuration space that is integrated over and depends upon the temperature through the Boltzmann

weight assigned to each configuration. Consequently, effective potentials that are designed as systematic approximations to the PMF should be expected to demonstrate similar state point dependence.

Considerable effort has been devoted to studying the temperature dependence of CG effective potentials. Several studies have demonstrated that, in some cases, CG potentials parameterized at a single temperature can provide an accurate description of atomistic structure over a wide range of temperatures. [116, 147, 178] However, in other cases CG potentials have demonstrated relatively little temperature transferability. [179] Moreover, Ghosh and Faller warned that, even if the target atomistic distribution functions demonstrate little temperature-dependence, effective potentials that are parameterized to reproduce the atomistic structure at one temperature may not accurately reproduce the atomistic structure at the second temperature. [117] Several studies have also proposed systematic methods for modifying effective potentials parameterized at one temperature for simulations at a second temperature. For instance, Krishna et al. adapted the Ferrenberg-Swendsen reweighting algorithm [180] to use configurations sampled from the canonical ensemble at one temperature to determine the MS-CG potential at a second temperature. [94] Alternatively, Müller-Plathe and coworkers employed an empirical analytic function to modify an effective potential for simulating ethylbenzene over a wide temperature range. [118]

Somewhat less progress has been achieved in adapting systematic coarse-graining methods to determine effective potentials for modeling multiple molecular systems. Some studies have employed simple mixing rules to determine effective potentials for binary mixtures from potentials parameterized for one component systems. [118, 181, 182] Allen and Rutledge have proposed a novel local-density-dependent adaptation of the IMC procedure to reproduce both the molecular structure and also the chemical potential of atomistic models for several different systems. [183, 184] Mullinax and Noid generalized the MS-CG variational principle for "extended ensembles," which are defined by combining equilibrium ensembles for multiple distinct systems. They then demonstrated that this approach systematically improved the transferability of potentials for modeling a variety of liquid mixtures over a wide range of compositions. [162] Several other studies have attempted to minimize the "environment-mediated" contribution to effective pair potentials in order to obtain a more transferable direct potential. Wang, Voth, and coworkers have demonstrated that such potentials can provide surprisingly good transferability for ionic liquids. [178] Villa, Peter, and van der Vegt

have employed a similar strategy to determine effective site-site pair potentials by subtracting the environment-mediated contribution from the corresponding ppmf. [185, 186] Recently, they have applied this approach to determine implicit solvent benzene models for a range of benzene-water mixtures. [187] They demonstrated that the resulting potentials correctly incorporate hydrophobic interactions over a range of concentrations and that, moreover, these potentials perform significantly better than potentials parameterized for neat benzene.

A third aspect of transferability addresses the utility of effective potentials in different ensembles. The present chapter has been entirely developed within the canonical ensemble. However, effective potentials are frequently employed in simulations sampling, e.g., the isothermal-isobaric ensemble. Most commonly, CG simulations at constant NPT are performed by coupling the system to a barostat using the standard equations for the virial. [85] The resulting pressure of the CG model is then typically much too large. [176] This is not entirely surprising and is consistent with Weeks-Chandler-Andersen (WCA) theory, which implies that the local structure reported in, e.g., RDFs, primarily reflects the short-ranged, rapidly varying repulsive contributions to the potential and is relatively insensitive to the slowly varying long-ranged interactions that play an important role in stabilizing condensed phases. [188] Consequently, CG potentials are frequently modified so that, when using standard (atomistic) expressions for the virial, the potential generates the correct average virial. This modification typically takes one of two forms:

1. CG potentials that are parameterized via iterative structure-motivated methods (e.g., IMC, IBI) to reproduce structural distribution functions are often modified by the addition of a linear ramp potential: $U_{ramp}(x) = a(x - x_0)$. [107, 123, 179] This modification introduces an additional term in the virial. The slope of the ramp potential can then be varied so that the CG model generates the correct average virial. Because the ramp potential varies slowly, it has relatively little (negative) impact upon the local structure of the CG model (as reported by, e.g., RDFs), in accord with WCA theory.

2. CG potentials can also be adapted for NPT simulations by employing an additional virial constraint in the potential calculation. [128, 141, 142] This constraint then modifies the variational framework to determine the potential that best approximates the many-body PMF, subject to the constraint that the model generates the correct average virial when averaged over the configurations sampled from atomistic MD simu-

lations.

Although, both approaches have enjoyed some success, neither approach rigorously addresses the statistical thermodynamics of coarse-graining in the NPT ensemble. Recently Das and Andersen performed such an analysis. [189] A key insight in this work was the formulation of a consistency criterion not only for sampling configurations, but also for sampling volume fluctuations in the NPT ensemble. They demonstrated the intuitive result that, in the thermodynamic limit, the optimal effective potential should be the same in the NVT and NPT ensembles. However, an additional volume dependent contribution should be considered in the extended Lagrangian of the CG model. In some sense, this addresses the fact that the CG PMF is a function of volume and, consequently, the atomistic expression for the virial is no longer appropriate. This volume dependence is manifest not only in the configuration dependent contribution of the PMF (i.e., $-k_B T \ln z(\mathbf{R})$ in Eq. (15)) but also in the configuration-independent contribution to the PMF (i.e. *const* in the same equation). This analysis merits further consideration in future work.

A related issue is the issue of representability, i.e., the issue of which properties can possibly be accurately described with a CG model and an approximate potential. As detailed in this chapter, a variety of different avenues can be followed in deriving CG potentials and no simple approximate potential can be expected to accurately reproduce every detail (e.g., thermodynamics, structure, and kinetics) of an atomistic model. For instance, it is readily shown that the (unique) pair potential reproducing a given RDF will not reproduce thermodynamic properties such as the energy. [135] However, a more subtle consideration is the conclusion by Louis [190] that "an effective density dependent pair potential $v(r;\rho)$ cannot be properly interpreted without reference to the coarse-graining procedure by which it was derived." This is clearly illustrated by the analysis of Das and Andersen, showing that volume dependent contributions to the PMF should be addressed when attempting to reproduce the thermodynamic pressure with a CG model. [189] Along similar lines, Voth and coworkers have recently suggested a mechanism for addressing the significance of missing atomic degrees of freedom when considering CG observables. [160] Consequently, a major advance would be achieved if one could a priori determine which properties could possibly be reproduced by a CG model of a given form.

4.2. Mappings and Basis sets

CG mappings are most frequently determined by physico-chemical intuition for a particular system. CG sites are typically assigned to provide adequate resolution for describing functional groups that are believed to be important in a particular process. This approach is often particularly useful in the case that the mapping is relatively "fine" and each site corresponds to a relatively small number of atoms. In contrast, this approach is less straightforward when the CG mapping is more "coarse" and each site corresponds to, e.g., dozens or hundreds of atoms. A number of studies have considered the design of CG mappings in this latter case. For instance, very low resolution mappings have been designed for proteins and biomolecules based upon consideration of rigidity analysis, [191] preserving dynamic domains, [192, 193] or retaining spatial mass distributions. [194, 195]

Nevertheless, from the perspective of the author, the CG mapping remains quite underappreciated and poorly understood. The theory outlined above demonstrates that, subject to relatively few assumptions, for any atomistic model and associated mapping, there does exist a many-body PMF and a CG model with this potential will reproduce all of the structural properties of the underlying atomistic model. However, the PMF can never be attained and, instead, must be approximated with an incomplete basis set. The accuracy and transferability of the resulting potential then depend upon a subtle relationship between the mapping and the basis set. For a given mapping, the accuracy and transferability of the potential are fundamentally limited by the flexibility of the basis set. The accuracy and transferability of the potential can only be improved by expanding the basis set. Typically, molecular mechanics potentials are employed for their convenience and efficiency in CG simulations. It is not clear what will be the most useful potentials for expanding the basis set. Several recent studies have considered Stillinger-Weber [196] type potentials, [77, 160] generalizations of Gay-Berne potentials, [76, 197] or other noncentral potentials. [78, 198, 199] However, much work remains to be done in identifying a better basis that allow accurate and transferable potentials to be efficiently simulated. Conversely, for a given basis set, the accuracy and transferability of the resulting model strongly depends upon the CG mapping. For instance, Rühle et al. demonstrated that a simple molecular mechanics basis cannot accurately capture atomistic forces that appear as noncentral forces in the CG model. [120] Moreover, they demonstrated that these errors led to a poor description of the conformers

sampled by a CG hexane model. In addition, Mullinax and Noid recently demonstrated the somewhat counterintuitive result that the accuracy and transferability of CG potentials can actually decrease when increasing the resolution of the CG mapping. [162]

4.3. Predictive capability

Finally, a third major challenge addresses the capability to predict the accuracy and transferability of CG models. Very frequently, CG potentials are only validated when their results are compared with results from high resolution models, most often for the system that was used in parameterizing the potential. Relatively little progress has been achieved in attempting to predict the accuracy of a CG model for a particular system. Recently, Chaimovich and Shell suggested that the relative entropy may allow predictions of the accuracy of a CG model, while using information extracted from the atomistic model, i.e., before actually performing CG simulations. [200] Alternatively, it may be possible to formulate self-consistency criterion when determining MS-CG potentials. The development of a priori estimates of the accuracy and transferability of CG models will represent a major advance.

5. CONCLUSIONS

In recent years, tremendous progress has been achieved in the development of new theories and computational methods for systematic coarse-graining. This chapter reviews a formal framework for considering these advances and then attempts to provide a unified discussion of both structure-motivated and also force-motivated approaches. It is shown that a generalization of the Yvon-Born-Green theory provides a conceptual bridge between the two approaches. The chapter also discusses some of the most outstanding challenges facing the field and reviews some of the progress made to date in addressing these challenges. It is anticipated that these challenges may well be overcome in the near future and it is hoped that the unified framework presented in this chapter may play some small role in stimulating these advances.

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References

[1] Allen, M. P. and Tildesley, D. P. (1987) Computer Simulation of Liquids. Oxford Press.

[2] Frenkel, D. and Smit, B. (2002) Understanding Molecular Simulation: From Algorithms to Applications. Academic Press, second edn.

[3] Cornell, W. D., Cieplak, P., Bayly, C. I., Gould, I. R., Merz, K. M., Ferguson, D. M., Spellmeyer, D. C., Fox, T., Caldwell, J. W., and Kollman, P. A. (1995) A 2nd generation force-field for the simulation of proteins, nucleic-acids, and organic-molecules. J. Am. Chem. Soc., **117**, 5179–97.

[4] MacKerell, A. D., et al. (1998) All-atom empirical potential for molecular modeling and dynamics studies of proteins. J. Phys. Chem. B, **102**, 3586 – 3616.

[5] Jorgensen, W. L., Maxwell, D. S., and Tirado-Rives, J. (1996) Development and testing of the OPLS All-Atom force field on conformational energetics and properties of organic liquids. J. Am. Chem. Soc., **118**, 11225–36.

[6] Mervis, J. (2001) Advanced computing - nsf launches teragrid for academic research. Science, **293**, 1235–6.

[7] Reed, D. A. (2003) Grids, the teragrid, and beyond. Computer, **36**, 62 – 68.

[8] Brooks, B. R., Bruccoleri, R. E., Olafson, B. D., States, D. J., Swaminathan, S., and Karplus, M. (1983) CHARMM - A program for macromolecular energy, minimization, and dynamics calculations. J. Comp. Chem., **4**, 187–217.

[9] Plimpton, S. (1995) Fast parallel algorithms for short-range molecular-dynamics. J. Comp. Phys., **117**, 1 – 19.

[10] Case, D. A., T. E. Cheatham, I., Darden, T., Gohlke, H., Luo, R., K. M. Merz, J., Onufriev, A., Simmerling, C., Wang, B., and Woods, R. (2005) The Amber biomolecular simulation programs. J. Comp. Chem., **26**, 1668–88.

[11] Christen, M., et al. (2005) The gromos software for biomolecular simulation: Gromos05. J. Comp. Chem., **26**, 1719–1751.

[12] Hess, B., Kutzner, C., van der Spoel, D., and Lindahl, E. (2008) Gromacs 4: Algorithms for highly efficient, load-balanced, and scalable molecular simulation. J. Chem. Theor. Comp., **4**, 435–447.

[13] Duan, Y. and Kollman, P. A. (1998) Pathways to a protein folding intermediate observed in a 1-microsecond simulation in aqueous solution. Science, **282**, 740–743.

[14] Gumbart, J., Trabuco, L. G., Schreiner, E., Villa, E., and Schulten, K. (2009) Regulation of the protein-conducting channel by a bound ribosome. Structure, **17**, 1453–1464.

[15] Anderson, P. W. (1972) More is different. Science, **177**, 393–6.

[16] Boon, J.-P. and Yip, S. (1992) Molecular Hydrodynamics. Dover Publications.

[17] Callen, H. B. (1985) Thermodynamics and an Introduction to Thermostatistics. Wiley.

[18] Clementi, C. (2008) Coarse-grained models of protein folding: Toy models or predictive tools? Curr. Opin. Struc. Biol., **18**, 10–15.

[19] Kirkwood, J. G. (1935) Statistical mechanics of fluid mixtures. J. Chem. Phys., **3**, 300–313.

[20] Kirkwood, J. G. (1946) The statistical mechanical theory of transport processes .1. general theory. J. Chem. Phys., **14**, 180–201.

[21] Irving, J. H. and Kirkwood, J. G. (1950) The statistical mechanical theory of transport processes .4. the equations of hydrodynamics. J. Chem. Phys., **18**, 817–829.

[22] Silbermann, J., Klapp, S. H. L., Shoen, M., Channamsetty, N., Block, H., and Gubbins, K. E. (2006) Mesoscale modeling of complex binary fluid mixtures: Towards an atomistic foundation of effective potentials. J. Chem. Phys., **124**, 074105.

[23] Levitt, M. and Warshel, A. (1975) Computer simulation of protein folding. Nature, **253**, 694–8.

[24] Marrink, S. J., de Vries, A. H., and Mark, A. E. (2004) Coarse grained model for semiquantitative lipid simulations. J. Phys. Chem. B, **108**, 750 – 760.

[25] ten Wolde, P. R. and Chandler, D. (2002) Drying-induced hydrophobic polymer collapse. Proc. Natl. Acad. Sci. USA, **99**, 6539–6543.

[26] Everaers, R., Sukumaran, S. K., Grest, G. S., Svaneborg, C., Sivasubramanian, A., and Kremer, K. (2004) Rheology and microscopic topology of entangled polymeric liquids. Science, **303**, 823–826.

[27] Reynwar, B. J., Illya, G., Harmandaris, V. A., Muller, M. M., Kremer, K., and Deserno, M. (2007) Aggregation and vesiculation of membrane proteins by curvature-mediated interactions. Nature, **447**, 461–464.

[28] Thirumalai, D. and Klimov, D. K. (1999) Deciphering the timescales and mechanisms of protein folding using minimal off-lattice models. Curr. Opin. Struc. Biol., **9**, 197–207.

[29] Hagler, A. T. and Honig, B. (1978) On the formation of protein tertiary structure on a computer. Proc. Natl. Acad. Sci. USA, **75**, 554–8.

[30] Shelley, J. C., Shelley, M. Y., Reeder, R. C., Bandyopadhyay, S., and Klein, M. L. (2001) A coarse grain model for phospholipid simulation. J. Phys. Chem. B, **105**, 4464–70.

[31] Nielsen, S. O., Lopez, C. F., Srinivas, G., and Klein, M. L. (2003) A coarse grain model for n-alkanes parameterized from surface tension data. J. Chem. Phys., **119**, 7043–49.

[32] Shinoda, W., Devane, R., and Klein, M. L. (2007) Multi-property fitting and parameterization of a coarse grained model for aqueous surfactants. Mol. Sim., **33**, 27–36.

[33] Khurana, E., DeVane, R. H., Kohlmeyer, A., and Klein, M. L. (2008) Probing peptide nanotube self-assembly at a liquid-liquid interface with coarse-grained molecular dynamics. Nanoletters, **8**, 3626–30.

[34] DeVane, R., Shinoda, W., Moore, P. B., and Klein, M. L. (2009) Transferable coarse grain nonbonded interaction model for amino acids. J. Chem. Theor. Comp., **5**, 2115–2124.

[35] DeVane, R., Klein, M. L., Chiu, C. C., Nielsen, S. O., Shinoda, W., and Moore, P. B. (2010) Coarse-grained potential models for phenyl-based molecules: I. parametrization using experimental data. J. Phys. Chem. B, **114**, 6386–6393.

[36] Chiu, C. C., DeVane, R., Klein, M. L., Shinoda, W., Moore, P. B., and Nielsen, S. O. (2010) Coarse-grained potential models for phenyl-based molecules: Ii. application to fullerenes. J. Phys. Chem. B, **114**, 6394–6400.

[37] Marrink, S. J., Risselada, H. J., Yefimov, S., Tieleman, D. P., and de Vries, A. H. (2007) The MARTINI force field: Coarse grained model for biomolecular simulations. J. Phys. Chem. B, **111**, 7812–7824.

[38] Monticelli, L., Kandasamy, S. K., Periole, X., Larson, R. G., Tieleman, D. P., and Marrink, S.-J. (2008) The MARTINI coarse-grained force field: Extension to proteins. J. Chem. Theor. Comp., **4**, 819–34.

[39] Lopez, C. A., Rzepiela, A. J., de Vries, A. H., Dijkhuizen, L., Hunenberger, P. H., and Marrink, S. J. (2009) Martini coarse-grained force field: Extension to carbohydrates. J. Chem. Theor. Comp., **5**, 3195–3210.

[40] Periole, X., Cavalli, M., Marrink, S. J., and Ceruso, M. A. (2009) Combining an elastic network with a coarse-grained molecular force field: Structure, dynamics, and intermolecular recognition. J. Chem. Theor. Comp., **5**, 2531–2543.

[41] Lee, H., de Vries, A. H., Marrink, S. J., and Pastor, R. W. (2009) A coarse-grained model for polyethylene oxide and polyethylene glycol: Conformation and hydrodynamics. J. Phys. Chem. B, **113**, 13186–13194.

[42] Partial special issue of J. Chem. Theory Comp. Volume 2, Issue 3 May 2006.

[43] Phys. Chem. Chem. Phys. Themed issue on coarse-grained modeling soft condensed matter. Volume 11, Issue 12 2009.

[44] Soft Matter themed issue on modeling soft matter. Volume 2, Issue 22 2009.

[45] Faraday Discussions on multiscale simulation of soft matter systems Volume 144 2010.

[46] Voth, G. A. (ed.) (2009) Coarse-graining of condensed phase and biomolecular systems. CRC Press.

[47] Ayton, G. S., Noid, W. G., and Voth, G. A. (2007) Multiscale modeling of biomolecular systems: In serial and in parallel. Curr. Opin. Struc. Biol., **17**, 192–8.

[48] Peter, C. and Kremer, K. (2009) Multiscale simulation of soft matter systems - from the atomistic to the coarse-grained level and back. Soft Matter, **5**, 4357–4366.

[49] Murtola, T., Bunker, A., Vattulainen, I., Deserno, M., and Karttunen, M. (2009) Multiscale modeling of emergent materials: Biological and soft matter. Phys. Chem. Chem. Phys., **11**, 1869–92.

[50] Kroger, M. (2004) Simple models for complex nonequilibrium fluids. Phys. Rep., **390**, 453–551.

[51] E, W., Engquist, B., Li, X. T., Ren, W. Q., and Vanden-Eijnden, E. (2007) Heterogeneous multiscale methods: A review. Comm. Comp. Phys., **2**, 367–450.

[52] Head-Gordon, T. and Brown, S. (2003) Minimalist models for protein folding and design. Curr. Opin. Struc. Biol., **13**, 160–167.

[53] Bahar, I. and Rader, A. (2005) Coarse-grained normal mode analysis in structural biology. Curr. Opin. Struc. Biol., **15**, 586–592.

[54] Tozzini, V. (2005) Coarse-grained models for proteins. Curr. Opin. Struc. Biol., **15**, 144 – 150.

[55] Bond, P. J., Holyoake, J., Ivetac, A., Khalid, S., and Sansom, M. S. P. (2007) Coarse-grained molecular dynamics simulations of membrane proteins and peptides. J. Struct. Biol., **157**, 593–605.

[56] Liwo, A., Czaplewski, C., Oldziej, S., and Scheraga, H. A. (2008) Computational techniques for efficient conformational sampling of proteins. Curr. Opin. Struc. Biol., **18**, 134–139.

[57] Hills, R. D. and Brooks, C. L. (2009) Insights from coarse-grained go models for protein folding and dynamics. Int. J. Mol. Sci., **10**, 889–905.

[58] Kuznetsov, I. B. (2009) Simplified computational methods for the analysis of protein flexibility. Curr. Prot. Pept. Sci., **10**, 607–613.

[59] Eom, K., Yoon, G., Kim, J. I., and Na, S. (2010) Coarse-grained elastic models of protein structures for understanding their mechanics and dynamics. J. Comp. Theor. Nanosci., **7**, 1210–1226.

[60] Tozzini, V. (2010) Multiscale modeling of proteins. Acc. Chem. Res., **43**, 220–230.

[61] Bahar, I., Lezon, T. R., Yang, L. W., and Eyal, E. (2010) Global dynamics of proteins: Bridging between structure and function. Ann. Rev. Biochem., **39**, 23–42.

[62] Scott, H. L. (2002) Modeling the lipid component of membranes. Curr. Opin. Struc. Biol., **12**, 495 – 502.

[63] Brannigan, G., Lin, L. C.-L., and Brown, F. L. H. (2006) Implicit solvent simulation models for biomembranes. Eur. Biophys. J., **35**, 104–24.

[64] Muller, M., Katsov, K., and Schick, M. (2006) Biological and synthetic membranes: What can be learned from a coarse-grained description? Phys. Rep., **434**, 113–176.

[65] Venturoli, M., Sperotto, M. M., Kranenburg, M., and Smit, B. (2006) Mesoscopic models of biological membranes. Phys. Rep., **437**, 1–54.

[66] Brown, F. L. H. (2008) Elastic modeling of biomembranes and lipid bilayers. Annu. Rev. Phys. Chem., **59**, 685–712.

[67] Bennun, S. V., Hoopes, M. I., Xing, C. Y., and Faller, R. (2009) Coarse-grained modeling of lipids. Chem. Phys. Lipids, **159**, 59–66.

[68] Noguchi, H. (2009) Membrane simulation models from nanometer to micrometer scale. J. Phys. Soc. Jpn., **78**.

[69] Pandit, S. A. and Scott, H. L. (2009) Multiscale simulations of heterogeneous model membranes. Biochim. Biophys. Act. - Biomembranes, **1788**, 136–148.

[70] Deserno, M. (2009) Mesoscopic membrane physics: Concepts, simulations, and selected applications. Macromol. Rapid Comm., **30**, 752–771.

[71] Glotzer, S. C. and Paul, W. (2002) Molecular and mesoscale simulation methods for polymer materials. Ann. Rev. Mat. Res., **32**, 401–436.

[72] Paul, W. and Smith, G. D. (2004) Structure and dynamics of amorphous polymers: computer simulations compared to experiment and theory. Rep. Prog. Phys., **67**, 1117–1185.

[73] Guenza, M. G. (2008) Theoretical models for bridging timescales in polymer dynamics. J. Phys.: Condens. Matter, **20**.

[74] Ganesan, V., Ellison, C. J., and Pryamitsyn, V. (2010) Mean-field models of structure and dispersion of polymer-nanoparticle mixtures. Soft Matter, **6**, 4010–4025.

[75] Buchete, N. V., Straub, J. E., and Thirumalai, D. (2004) Development of novel statistical potentials for protein fold recognition. Curr. Opin. Struc. Biol., **14**, 225–32.

[76] Babadi, M., Everaers, R., and Ejtehadi, M. R. (2006) Coarse-grained interaction potentials for anisotropic molecules. J. Chem. Phys., **124**.

[77] Molinero, V. and Moore, E. B. (2009) Water modeled as an intermediate element between carbon and silicon. J. Phys. Chem. B, **113**, 4008–4016.

[78] Alemani, D., Collu, F., Cascella, M., and Dal Peraro, M. (2010) A nonradial coarse-grained potential for proteins produces naturally stable secondary structure elements. J. Chem. Theor. Comp., **6**, 315–324.

[79] Murtola, T., Karttunen, M., and Vattulainen, I. (2009) Systematic coarse graining from structure using internal states: Application to phospholipid/cholesterol bilayer. J. Chem. Phys., **131**, 055101.

[80] Noid, W. G., Chu, J.-W., Ayton, G. S., Krishna, V., Izvekov, S., Voth, G. A., Das, A., and Andersen, H. C. (2008) The multiscale coarse-graining method. I. A rigorous bridge between atomistic and coarse-grained models. J. Chem. Phys., **128**, 244114.

[81] Hill, T. L. (1987) An introduction to statistical thermodynamics. Dover reprint.

[82] Shi, Z., Chen, K., Liu, Z., and Kallenbach, N. R. (2006) Conformation of the backbone in unfolded proteins. Chem. Rev., **106**, 1877–1897.

[83] Praprotnik, M., Delle Site, L., and Kremer, K. (2008) Multiscale simulation of soft matter: From scale bridging to adaptive resolution. Annu. Rev. Phys. Chem., **59**, 545–571.

[84] Ensing, B., Nielsen, S. O., Moore, P. B., Klein, M. L., and Parrinello, M. (2007) Energy conservation in adaptive hybrid atomistic/coarse-grain molecular dynamics. J. Chem. Theor. Comp., **3**, 1100–1105.

[85] Hansen, J.-P. and McDonald, I. R. (1990) Theory of Simple Liquids. Academic Press, 2 edn.

[86] Liwo, A., Oldziej, S., Pincus, M. R., Wawak, R. J., Rackovsky, S., and Scheraga, H. A. (1997) A united-residue force field for off-lattice protein-structure simulations. 1. Functional forms and parameters of long-range side-chain interaction potentials from protein crystal data. J. Comp. Chem., **18**, 849–73.

[87] van Kampen, N. G. (2007) Stochastic Processes in Physics and Chemistry. North-Holland, 3 edn.

[88] Zwanzig, R. (1960) Ensemble method in the theory of irreversibility. J. Chem. Phys., **33**, 1338–1341.

[89] Mori, H. (1965) Transport collective motion and brownian motion. Prog. Theor. Phys., **33**, 423–55.

[90] Baschnagel, J., et al. (2000) Bridging the gap between atomistic and coarse-grained models of polymers: Status and perspectives. Adv. Poly. Sci., **152**, 41–156.

[91] Shell, M. S. (2008) The relative entropy is fundamental to multiscale and inverse thermodynamic problems. J. Chem. Phys., **129**.

[92] Bender, C. M. and Orszag, S. A. (1999) Advanced Mathematical Methods for Scientists and Engineers: Asymptotic Methods and Perturbation Theory. Springer.

[93] Kullback, S. and Leibler, R. A. (1951) On information and sufficiency. Ann. Math. Stat., **22**, 79–86.

[94] Krishna, V., Noid, W. G., and Voth, G. A. (2009) The multiscale coarse-graining method. iv. transferring coarse-grained potentials between temperatures. J. Chem. Phys., **131**, 024103.

[95] Chaimovich, A. and Shell, M. S. (2011) Coarse-graining errors and numerical optimization using a relative entropy framework. J. Chem. Phys., **134**, 094112.

[96] Rudzinski, J. F. and Noid, W. G. (2011) Coarse-graining entropy, forces, and structures. J. Chem. Phys., **135**, 214101.

[97] Larini, L. and Krishna, V. (2011) A generalized mean field theory of coarse-graining. J. Chem. Phys., **135**, 124103.

[98] Evans, R. (1979) Nature of the liquid-vapor interface and other topics in the statistical-mechanics of nonuniform, classical fluids. Adv. Phys., **28**, 143–200.

[99] Evans, R. (1990) Comment on reverse monte carlo simulation. Mol. Sim., **4**, 409–411.

[100] Tschop, W., Kremer, K., Batoulis, J., Burger, T., and Hahn, O. (1998) Simulation of polymer melts. i. coarse-graining procedure for polycarbonates. Acta Poly., **49**, 61–74.

[101] Praprotnik, M., Junghans, C., Delle Site, L., and Kremer, K. (2008) Simulation approaches to soft matter: Generic statistical properties vs. chemical details. Comp. Phys. Comm., **179**, 51–60.

[102] Peter, C. and Kremer, K. (2009) Multiscale simulation of soft matter systems - from the atomistic to the coarse-grained level and back. Soft Matter, **5**, 4357–4366.

[103] Harmandaris, V. A., Adhikari, N. P., van der Vegt, N. F. A., and Kremer, K. (2006) Hierarchical modeling of polystyrene: From atomistic to coarse-grained simulations. Macromolecules, **39**, 6708–6719.

[104] Peter, C., Delle Site, L., and Kremer, K. (2008) Classical simulations from the atomistic to the mesoscale and back: coarse graining an azobenzene liquid crystal. Soft Matter, **4**, 859–869.

[105] Hill, T. L. (1987) Statistical Mechanics: Principles and selected applications. Dover reprint.

[106] Noid, W. G., Chu, J.-W., Ayton, G. S., and Voth, G. A. (2007) Multiscale coarse-graining and structural correlations: Connections to liquid state theory. J. Phys. Chem. B, **111**, 4116–27.

[107] Müller-Plathe, F. (2002) Coarse-graining in polymer simulation: From the atomistic to the mesoscopic scale and back. ChemPhysChem, **3**, 754 – 769.

[108] Schommers, W. (1983) Pair potentials in disordered many-particle systems: A study of liquid gallium. Phys. Rev. A, **28**, 3599–3605.

[109] Soper, A. K. (1996) Empirical potential monte carlo simulation of fluid structure. Chem. Phys., **202**, 295–306.

[110] Reatto, L., Levesque, D., and Weis, J. J. (1986) Iterative predictor-corrector method for extraction of the pair interaction from structural data for dense classical liquids. Phys. Rev. A, **33**, 3451–3465.

[111] McGreevy, R. L. and Pusztai, L. (1988) Reverse Monte Carlo simulation: A new technique for the determination of disordered structures. Mol. Sim., **1**, 359–67.

[112] Keen, D. A. and McGreevy, R. L. (1990) Structural modelling of glasses using reverse Monte Carlo simulation. Nature, **344**, 423–5.

[113] Reith, D., Putz, M., and Muller-Plathe, F. (2003) Deriving effective mesoscale potentials from atomistic simulations. J. Comp. Chem., **24**, 1624–1636.

[114] Milano, G. and Müller-Plathe, F. (2005) Mapping atomistic simulations to mesoscopic models: A systematic coarse-graining procedure for vinyl polymer chains. J. Phys. Chem. B, **109**, 18609–19.

[115] Sun, Q. and Faller, R. (2006) Crossover from unentangled to entangled dynamics in a systematically coarse-grained polystyrene melt. Macromolecules, **39**, 812–820.

[116] Vettorel, T. and Meyer, H. (2006) Coarse graining of short polyethylene chains for studying polymer crystallization. J. Chem. Theor. Comp., **2**, 616–629.

[117] Ghosh, J. and Faller, R. (2007) State point dependence of systematically coarse-grained potentials. Mol. Sim., **33**, 759–767.

[118] Qian, H.-J., Carbone, P., Chen, X., Karimi-Varzaneh, H. A., Liew, C. C., and Müller-Plathe, F. (2008) Temperature-transferable coarse-grained potentials for ethylbenzene, polystyrene and their mixtures. Macromolecules, **41**, 9919–29.

[119] Fischer, J., Paschek, D., Gieger, A., and Sadowski, G. (2008) Modeling of aqueous poly(oxyethylene) solutions. 2. Mesoscale simulations. J. Phys. Chem. B, **112**, 13561–71.

[120] Ruhle, V., Junghans, C., Lukyanov, A., Kremer, K., and Andrienko, D. (2009) Versatile object-oriented toolkit for coarse-graining applications. J. Chem. Theor. Comp., **5**, 3211–3223.

[121] Wang, H., Junghans, C., and Kremer, K. (2009) Comparative atomistic and coarse-grained study of water: What do we lose by coarse-graining? Eur. Phys. J. E, **28**, 221–229.

[122] Megariotis, G., Vyrkou, A., Leygue, A., and Theodorou, D. N. (2010) Systematic coarse graining of 4-cyano-4-pentylbiphenyl, In press Ind. Eng. Chem. Res. ASAP.

[123] Reith, D., Pütz, M., and Müller-Plathe, F. (2003) Deriving effective mesoscale potentials from atomistic simulations. J. Comp. Chem., **24**, 1624–36.

[124] Lyubartsev, A. P. and Laaksonen, A. (1995) Calculation of effective interaction potentials from radial distribution functions: A reverse Monte Carlo approach. Phys. Rev. E, **52**, 3730–37.

[125] Lyubartsev, A., Mirzoev, A., Chen, L. J., and Laaksonen, A. (2010) Systematic coarse-graining of molecular models by the newton inversion method. Faraday Disc., **144**, 43–56.

[126] Murtola, T., Falck, E., Patra, M., Karttunen, M., and Vattulainen, I. (2004) Coarse-grained model for phospholipid/cholesterol bilayer. J. Chem. Phys., **121**, 9156–65.

[127] Lyubartsev, A. P. (2005) Multiscale modeling of lipids and lipid bilayers. Eur. Biophys. J., **35**, 53–61.

[128] Murtola, T., Falck, E., Karttunen, M., and Vattulainen, I. (2007) Coarse-grained model for phospholipid/cholesterol bilayer employing inverse Monte Carlo with thermodynamic constraints. J. Chem. Phys., **121**, 075101.

[129] Lyubartsev, A. P. and Laaksonen, A. (1997) Osmotic and activity coefficients from effective potentials for hydrated ions. Phys. Rev. E, **55**, 5689–96.

[130] Lyubartsev, A. P., Karttunen, M., Vattulainen, I., and Laaksonen, A. (2003) On coarse-graining by the inverse Monte Carlo method: Dissipative particle dynamics simulations made to a precise tool in soft matter modeling. Soft Matter, **1**, 121–37.

[131] Isihara, A. (1968) Gibbs-bogoliubov inequality. J. Phys. A: Math., Nucl., Gen., **1**, 539–548.

[132] Henderson, R. L. (1974) A uniqueness theorem for fluid pair correlation functions. Phys. Lett. A, **49**, 197–8.

[133] Hohenberg, P. and Kohn, W. (1964) Inhomogeneous electron gas. Phys. Rev. B, **136**, 864–871.

[134] Mermin, N. D. (1965) Thermal properties of inhomogeneous electron gas. Phys. Rev. A, **137**, 1441–3.

[135] Johnson, M. E., Head-Gordon, T., and Louis, A. A. (2007) Representability problems for coarse-grained water potentials. J. Chem. Phys., **126**, 144509.

[136] Chayes, J. T., Chayes, L., and Lieb, E. H. (1984) The inverse problem in classical statistical mechanics. Comm. Math. Phys., **93**, 57–121.

[137] Chayes, J. T. and Chayes, L. (1984) On the validity of the inverse conjecture in classical density functional theory. J. Stat. Phys., **36**, 471–88.

[138] Savelyev, A. and Papoian, G. A. (2009) Molecular renormalization group coarse-graining of electrolyte solutions: Applications to aqueous NaCl and KCl. J. Phys. Chem. B, **113**, 7785–93.

[139] Savelyev, A. and Papoian, G. A. (2009) Molecular renormalization group coarse-graining of polymer chains: Applications to double-stranded DNA. Biophys. J., **96**, 4044–52.

[140] Izvekov, S. and Voth, G. A. (2005) A multiscale coarse-graining method for biomolecular systems. J. Phys. Chem. B, **109**, 2469 – 2473.

[141] Izvekov, S. and Voth, G. A. (2005) Multiscale coarse graining of liquid-state systems. J. Chem. Phys., **123**, 134105.

[142] Izvekov, S., Chung, P. W., and Rice, B. M. (2010) The multiscale coarse-graining method: Assessing its accuracy and introducing density dependent coarse-grain potentials. J. Chem. Phys., **133**, 064109.

[143] Ercolessi, F. and Adams, J. B. (1994) Interatomic potentials from first-principles calculations: The force-matching method. Europhys. Lett., **26**, 583.

[144] Izvekov, S., Parrinello, M., Burnham, C. J., and Voth, G. A. (2004) Effective force fields for condensed phase systems from *ab initio* molecular dynamics simulation: A new method for force-matching. J. Chem. Phys., **120**, 10896 – 10913.

[145] Wang, Y. T. and Voth, G. A. (2005) Unique spatial heterogeneity in ionic liquids. J. Am. Chem. Soc., **127**, 12192 – 12193.

[146] Wang, Y. T., Izvekov, S., Yan, T. Y., and Voth, G. A. (2006) Multiscale coarse-graining of ionic liquids. J. Phys. Chem. B, **110**, 3564 – 3575.

[147] Wang, Y. T., Feng, S. L., and Voth, G. A. (2009) Transferable coarse-grained models for ionic liquids. J. Chem. Theor. Comp., **5**, 1091–1098.

[148] Izvekov, S. and Voth, G. A. (2006) Multiscale coarse-graining of mixed phospholipid/cholesterol bilayers. J. Chem. Theor. Comp., **2**, 637 – 648.

[149] Izvekov, S. and Voth, G. A. (2009) Solvent-free lipid bilayer model using multiscale coarse graining. J. Phys. Chem. B, **113**, 4443–55.

[150] Izvekov, S., Violi, A., and Voth, G. A. (2005) Systematic coarse-graining of nanoparticle interactions in molecular dynamics simulation. J. Phys. Chem. B, **109**, 17019 – 17024.

[151] Liu, P., Izvekov, S., and Voth, G. A. (2007) Multiscale coarse-graining of monosaccharides. J. Phys. Chem. B, **111**, 11566–75.

[152] Wang, Y. T. and Voth, G. A. (2010) Molecular dynamics simulations of polyglutamine aggregation using solvent-free multiscale coarse-grained models. J. Phys. Chem. B, **114**, 8735–8743.

[153] Hills, R. D., Lu, L. Y., and Voth, G. A. (2010) Multiscale coarse-graining of the protein energy landscape. PLoS Comput. Biol., **6**, e1000827.

[154] Chorin, A. J., Hald, O. H., and Kupferman, R. (2000) Optimal prediction and the Mori-Zwanzig representation of irreversible processes. Proc. Natl. Acad. Sci. USA, **97**, 2968–73.

[155] Noid, W. G., Liu, P., Wang, Y. T., Chu, J.-W., Ayton, G. S., Izvekov, S., Andersen, H. C., and Voth, G. A. (2008) The multiscale coarse-graining method. II. Numerical implementation for molecular coarse-grained models. J. Chem. Phys., **128**, 244115.

[156] Mullinax, J. W. and Noid, W. G. (2009) A generalized Yvon-Born-Green theory for molecular systems. Phys. Rev. Lett., **103**, 198104.

[157] Mullinax, J. W. and Noid, W. G. (2010) A generalized Yvon-Born-Green theory for determining coarse-grained interaction potentials. J. Phys. Chem. C, **114**, 5661–74.

[158] Chorin, A. J. (2003) Conditional expectations and renormalization. Multiscale Model. Simul., **1**, 105–18.

[159] Chorin, A. J. and Hald, O. H. (2006) Stochastic Tools in Mathematics and Science. Springer.

[160] Larini, L., Lu, L. Y., and Voth, G. A. (2010) The multiscale coarse-graining method. vi. implementation of three-body coarse-grained potentials. J. Chem. Phys., **132**, 164107.

[161] Lu, L. and Voth, G. A. (2009) Systematic coarse-graining of a multicomponent lipid bilayer. J. Phys. Chem. B, **113**, 1501–10.

[162] Mullinax, J. W. and Noid, W. G. (2009) Extended ensemble approach for deriving transferable coarse-grained potentials. J. Chem. Phys., **131**, 104110.

[163] Lu, L. Y., Izvekov, S., Das, A., Andersen, H. C., and Voth, G. A. (2010) Efficient, regularized, and scalable algorithms for multiscale coarse-graining. J. Chem. Theor. Comp., **6**, 954–965.

[164] Press, W. H., Teukolsky, S. A., Vetterling, W. T., and Flannery, B. P. (1992) Numerical Recipes in FORTRAN: The art of scientific computing. Cambridge University Press.

[165] Demmel, J. W. (1997) Applied Numerical Linear Algebra. SIAM.

[166] Mullinax, J. W. and Noid, W. G. (2010) Reference state for the generalized Yvon-Born-Green theory: Application for a coarse-grained model of hydrophobic hydration. J. Chem. Phys., **133**, 124107.

[167] Gubbins, K. E. (1980) Structure of nonuniform molecular fluids - integrodifferential equations for the density-orientation profile. Chem. Phys. Lett., **76**, 329–32.

[168] Whittington, S. G. and Dunfield, L. G. (1973) A Born-Green-Yvon treatment of polymers with excluded volume. J. Phys. A: Math., Nucl., Gen., **6**, 484.

[169] Attard, P. (1995) Polymer Born-Green-Yvon equation with proper triplet superposition approximation. results for hard-sphere chains. J. Chem. Phys., **102**, 5411.

[170] Taylor, M. P. and Lipson, J. E. G. (1993) A site-site Born-Green-Yvon equation for hard sphere dimers. J. Chem. Phys., **100**, 518.

[171] Taylor, M. P. and Lipson, J. E. G. (1995) A Born-Green-Yvon equation for flexible chain-molecule fluids. I. General formalism and numerical results for short hard-sphere chains. J. Chem. Phys., **102**, 2118.

[172] Taylor, M. P. and Lipson, J. E. G. (1995) A BornGreenYvon equation for flexible chain-molecule fluids. I. Applications to hard-sphere polymers. J. Chem. Phys., **102**, 6272.

[173] Katsov, K. and Weeks, J. D. (2001) On the mean field treatment of attractive interactions in nonuniform simple fluids. J. Phys. Chem. B, **105**, 6738–6744.

[174] Chen, Y. G., Kaur, C., and Weeks, J. D. (2004) Connecting systems with short and long ranged interactions: Local molecular field theory for ionic fluids. J. Phys. Chem. B, **108**, 19874–19884.

[175] Rodgers, J. M., Kaur, C., Chen, Y. G., and Weeks, J. D. (2006) Attraction between like-charged walls: Short-ranged simulations using local molecular field theory. Phys. Rev. Lett., **97**.

[176] Mullinax, J. W. and Noid, W. G. (2010) Recovering physical potentials from a model protein databank. Proc. Natl. Acad. Sci. USA, **107**, 19867.

[177] Cho, H. M. and Chu, J. W. (2009) Inversion of radial distribution functions to pair forces by solving the yvon-born-green equation iteratively. J. Chem. Phys., **131**, 134107.

[178] Wang, Y. T., Noid, W. G., Liu, P., and Voth, G. A. (2009) Effective force coarse-graining. Phys. Chem. Chem. Phys., **11**, 2002–2015.

[179] Carbone, P., Varzaneh, H. A. K., Chen, X., and Müller-Plathe, F. (2008) Transferability of coarse-grained force fields: The polymer case. J. Chem. Phys., **128**, 064904.

[180] Ferrenberg, A. M. and Swendsen, R. H. (1988) New monte carlo technique for studying phase transitions. Phys. Rev. Lett., **61**.

[181] Mognetti, B. M., Oettel, M., Yelash, L., Virnau, P., Paul, W., and Binder, K. (2008) Spherically averaged versus angle-dependent interactions in quadrupolar fluids. Phys. Rev. E, **77**, 041506.

[182] Mognetti, B. M., Virnau, P., Yelash, L., Paul, W., Binder, K., Müller, M., and MacDowell, L. G. (2009) Coarse-grained models for fluids and their mixtures: Comparison of Monte Carlo studies of their phase behavior with perturbation theory and experiment. J. Chem. Phys., **130**, 044101.

[183] Allen, E. C. and Rutledge, G. C. (2008) A novel algorithm for creating coarse-grained, density dependent implicit solvent models. J. Chem. Phys., **128**, 154115.

[184] Allen, E. C. and Rutledge, G. C. (2009) Evaluating the transferability of coarse-grained, density-dependent implicit solvent models to mixtures and chains. J. Chem. Phys., **130**, 034904.

[185] Villa, A., van der Vegt, N. F. A., and Peter, C. (2009) Self-assembling dipeptides: including solvent degrees of freedom in a coarse-grained model. Phys. Chem. Chem. Phys., **11**, 2068–2076.

[186] Villa, A., Peter, C., and van der Vegt, N. F. A. (2009) Self-assembling dipeptides: conformational sampling in solvent-free coarse-grained simulation. Phys. Chem. Chem. Phys., **11**, 2077–2086.

[187] Villa, A., Peter, C., and van der Vegt, N. F. A. (2010) Transferability of nonbonded interaction potentials for coarse-grained simulations: Benzene in water. J. Chem. Theor. Comp., **6**, 2434–2444.

[188] Andersen, H. C., Chandler, D., and Weeks, J. D. (1976) Roles of repulsive and attractive forces in liquids : The equilibrium theory of classical fluids. Adv. Chem. Phys., **34**, 105.

[189] Das, A. and Andersen, H. C. (2010) The multiscale coarse-graining method. V. Isothermal-isobaric ensemble. J. Chem. Phys., **132**, 164106.

[190] Louis, A. A. (2002) Beware of density dependent pair potentials. J. Phys.: Condens. Matter, **14**, 9187–206.

[191] Gohlke, H. and Thorpe, M. F. (2006) A natural coarse graining for simulating large biomolecular motion. Biophys. J., **91**, 2115–20.

[192] Zhang, Z. Y., Lu, L. Y., Noid, W. G., Krishna, V., Pfaendtner, J., and Voth, G. A. (2008) A systematic methodology for defining coarse-grained sites in large biomolecules. Biophys. J., **95**, 5073–5083.

[193] Zhang, Z. Y., Pfaendtner, J., Grafmuller, A., and Voth, G. A. (2009) Defining coarse-grained representations of large biomolecules and biomolecular complexes from elastic network mod-

els. Biophys. J., **97**, 2327–2337.

[194] Arkhipov, A., Freddolino, P., and Schulten, K. (2006) Stability and dynamics of virus capsids described by coarse-grained modeling. Structure, **129**, 1767–77.

[195] Zhang, Z. and Voth, G. A. (2010) Coarse-grained representations of large biomolecular complexes from low-resolution structural data. J. Chem. Theor. Comp., **6**, 2990–3002.

[196] Stillinger, F. H. and Weber, T. A. (1985) Computer-simulation of local order in condensed phases of silicon. Phys. Rev. B, **31**, 5262–5271.

[197] Morriss-Andrews, A., Rottler, J., and Plotkin, S. S. (2010) A systematically coarse-grained model for dna and its predictions for persistence length, stacking, twist, and chirality. J. Chem. Phys., **132**, 035105.

[198] Liwo, A., Czaplewski, C., Pillardy, J., and Scheraga, H. A. (2001) Cumulant-based expressions for the multibody terms for the correlation between local and electrostatic interactions in the united-residue force field. J. Chem. Phys., **115**, 2323 – 2347.

[199] Liwo, A., Kazmierkiewicz, R., Czaplewski, C., Groth, M., Oldziej, S., Wawak, R. J., Rackovsky, S., Pincus, M. R., and Scheraga, H. A. (1998) United-residue force field for off-lattice protein-structure simulations: III. Origin of backbone hydrogen-bonding cooperativity in united-residue potentials. J. Comp. Chem., **19**, 259 – 276.

[200] Chaimovich, A. and Shell, M. S. (2010) Relative entropy as a universal metric for multiscale errors. Phys. Rev. E, **81**.